

Early Childhood Development and Structural Change

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Abstract

This paper presents a theory of how early childhood development shapes structural change. I begin by documenting two novel cross-country facts: parental childcare time devoted to young children (ages 0–4) and early childhood human capital outcomes both rise with economic development. These patterns accompany the rising employment share of High-Skill Services with development. I then construct a heterogeneous agents general equilibrium model in which early childhood development raises the productivity of human capital investments in subsequent stages of childhood, increasing the supply of skills and, in turn, driving structural change. The model integrates a multi-stage human capital production function with dynamic complementarity and endogenous labor supply into a standard structural change model with non-homothetic preferences and unbalanced sectoral TFP growth. The model matches patterns of time use, structural change, and the skill distribution in the postwar US, and rationalizes both novel cross-country facts. I show that the rise in parental childcare time devoted to young children accounts for 35% of the US shift toward High-Skill Services in general equilibrium. Early childhood development is therefore a quantitatively important driver of structural change.

JEL Codes: E24, J13, J22, J24, O11, O41

Key Words: Early Childhood Development, Structural Change, High-Skill Services, Time Allocation, Dynamic Complementarity, Skills

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1 Introduction

There is a large body of work documenting the importance of early childhood human capital for life-cycle outcomes, focusing *inter alia* on skill formation (Cunha et al., 2006; Cunha and Heckman, 2007; Cunha et al., 2010), health (García and Heckman, 2021; Case et al., 2005), non-cognitive skills (Heckman, 2012; Heckman et al., 2013; Cunha et al., 2010), and crime (Lochner, 2010). In contrast, the literature on the implications of early childhood development on aggregate macroeconomic outcomes is less developed.

In this paper, I address this gap by asking how early childhood development shapes structural change — the reallocation of economic activity across sectors. The explicit focus on the sectoral composition of aggregate output is warranted given the intimate link between growth and development on the one hand and structural change on the other.¹ In this paper, I pay special attention to the structure of the Services sector given limited industrialization opportunities (Rodrik, 2016) and the Services sector’s growing importance as a source of employment in both advanced and developing countries.² Narrowing cross-country income differences will require that this transition be accompanied by the simultaneous reallocation of labor into skill-intensive, high-productivity industries such as High-Skill Services that can sustain higher wages and better living standards.

To answer my research question, I build on the existing theories of structural change — namely income effects via non-homothetic preferences (Kongsamut et al., 2001) and relative price effects via unbalanced sectoral productivity growth (Ngai and Pissarides, 2007) — by endogenizing the supply of skills in the economy through early childhood development. This extension enables an assessment of the effects of early childhood on human capital, income, and sectoral output, all of which may interact to shape structural change. Central to my argument that early childhood human capital shapes the supply of skills in the economy is the notion of dynamic complementarity: higher levels of human capital in *earlier* stages of childhood development make investments in *later* stages of childhood development more productive (Cunha and Heckman, 2007). Thus, the effects of early investments on childhood human capital not only persist over the life-cycle, but they also amplify the incentives for human capital accumulation in subsequent stages of childhood development. Higher early childhood human capital is therefore key to generating a more skill-intensive workforce over time. This can impact both sectoral

¹A significant empirical literature studies this phenomenon (starting from Fuchs, 1968; Kuznets, 1957; Baumol, 1967).

²Between 1975 and 2010, all advanced and developing countries covered in the 10-sector database (excluding Zambia) experienced an increase in their Services employment share. Roughly half of these countries have had their Services employment share rise by more than 20 percentage points (Timmer et al., 2015). These include both rich countries such as France (24.3%) and Italy (23.5%) as well as less developed economies such as Botswana (27.0%), Brazil (26.2%) and Thailand (22.8%).

demand (via higher income and non-homothetic preferences) and sectoral supply (via varying sectoral skill intensities), consequently shaping structural change.

To motivate my theory, I document two novel cross-country empirical facts that coincide with the rise in the supply of skills and the High-Skill Services employment share with development (Katz and Murphy, 1992; Acemoglu and Autor, 2011; Buera and Kaboski, 2012; Buera et al., 2022). First, using harmonized time use data, I show that parents in richer countries tend to allocate more time to the childcare of young children (ages 0–4).³ I verify that this pattern is along the intensive margin (i.e. per child) and robust across mothers and fathers. Second, I use harmonized microdata (IPUMS-MICS) to show that the early childhood human capital outcomes of three- and four-year-olds increase with development.

In order to uncover and quantify the link between early childhood development and structural change, I proceed by building a heterogeneous agents general equilibrium model. The model integrates a dynastic overlapping generations setting with endogenous labor supply and human capital accumulation into an otherwise standard multi-sector model of growth and structural change. In my model, parents face a trade-off between investing time in their children’s human capital, which evolves according to a multi-stage human capital production technology à la Del Boca et al. (2014), and supplying their labor to earn income. Parents obtain a static utility benefit from spending time with their children, which enters their preferences as a luxury, and a dynamic benefit from improving the children’s future human capital and, consequently, their future utility. On the other hand, the opportunity cost of time allocated to childcare is time at work which allows parents to earn an income which they can then spend on output from three different sectors: Goods, High-Skill Services, and Low-Skill Services. On the production side, each sector has a representative producer that chooses how much skilled and unskilled labor to hire in order to maximize profits in a perfectly competitive environment. The sectors differ in the skill intensity of their technologies and the (exogenous) pace of TFP growth. When children grow up, they become parents to children of their own and their human capital is fixed for the remainder of their lives. Upon entry into the labor force, parents face a Roy-type occupational choice problem in which, conditional on wages and their human capital level, they choose whether to be skilled or unskilled workers.

³Parental time allocation is an important — if not, the most important — input for early childhood human capital. Using a structural model, Del Boca et al. (2014) find that parental time allocation from both parents is the most important input for the cognitive development of young children. The importance of parental time allocation falls with the age of the child at the expense of other inputs. Furthermore, Yum (2023) finds that both parental time allocation and other inputs are poor substitutes in early stages of childhood development. Other papers that stress the role of parental time allocation for early childhood human capital outcomes relative to other inputs include Francesconi and Heckman (2016), Bono et al. (2016), Bernal (2008) and Fiorini and Keane (2014).

The supply-side mechanics of the model are amplified by income effects in time allocation. As income grows, non-homothetic preferences induce parents to invest more time in their children. Greater parental time investment improves early childhood human capital and, through dynamic complementarity, generates a more skilled workforce. This raises the productive capacity of the skill-intensive (High-Skill) Services sector. Simultaneously, the traditional demand-side drivers of structural change are at play; the sectoral consumption aggregator is non-homothetic à la [Comin et al. \(2021\)](#) such that as households get richer, parents bias their consumption towards luxuries and away from necessities.

In general equilibrium, the distribution of human capital is key. On the one hand, the distribution of human capital shapes the household income distribution which, together with relative prices, drives aggregate demand for High-Skill Services. On the other hand, the human capital distribution has distinct implications for the aggregate supply of High-Skill Services given its higher skill intensity. These forces jointly determine the return to skills. Together with parental preferences, this return shapes how much forward-looking parents invest in their children's human capital. Accordingly, the model is able to account for the endogenous feedback effects between the supply- and demand-side drivers of structural change.⁴

I then discipline the model to match patterns of structural change, time use and the skill distribution in the postwar United States (US).⁵ The quantitative version of the model is used to perform four counterfactual exercises. The first exercise explores the implications of the initial TFP level (the model's key driver of income levels) on structural change toward High-Skill Services. The results show that initial TFP influences structural change by governing the *level* of sectoral expenditure shares. A 25 percent decline in initial TFP across all sectors⁶ effectively reduces household incomes, leading to a 7.4 percentage point decline in the initial High-Skill Services expenditure share. Both supply and demand forces ensure that subsequent generations also experience comparatively lower expenditure shares in High-Skill Services; in response to lower incomes, parents invest less time in their young children because the return to skill is comparatively lower and because they bias their time allocation less towards childcare. Simultaneously, demand for High-Skill Services falls because households have less income. Consequently, both the

⁴The seminal paper by [Katz and Murphy \(1992\)](#) argues that the rising wage premium in the US resulted from a rise in demand for skills due to skill-biased technical change. [Buera et al. \(2022\)](#) extend this view by relating the rise in demand for skills with structural change. I enrich these frameworks by endogenizing the feedback between the demand for skills on the one hand, and the supply of skills — assumed exogenous in these papers — on the other.

⁵In the quantitative analysis, I measure structural change using sectoral expenditure shares which map directly into the model's household demand system. This reallocation is reflected in sectoral employment and value added shares.

⁶For context, this is broadly equivalent to the US-Zimbabwe and US-South Africa aggregate TFP gaps in the 1960s according to the Penn World Tables ([Feenstra et al., 2015](#)).

supply of skills and the High-Skill Services expenditure share are endogenously reduced relative to the baseline for subsequent generations.

The second and third counterfactual exercises quantify the role of early childhood development in shaping structural change. Specifically, the second counterfactual exercise measures the effect of the intergenerational rise in the time that parents allocate to the childcare of young children on US structural change. Fixing the share of time allocated to childcare of young children (ages 0–4) to the initial generation’s weighted average across all subsequent generations, I show that in partial equilibrium (i.e. fixing relative prices and wages to their baseline values) the rise in parental time allocated to childcare of young children explains 16.1 percentage points of the observed 25.1 percentage point increase in the High-Skill Services share of economic activity in the postwar US. This is equivalent to 64.3% of structural change toward High-Skill Services in the US. In general equilibrium (i.e. allowing relative prices and wages to clear markets), the contribution of rising parental childcare time declines to 35.0%. The results reveal that early childhood development improves the quality of human capital in the economy. This in turn amplifies the reallocation toward High-Skill Services by increasing demand for High-Skill Services and expanding the supply of skilled labor used intensively in their production.

The third exercise evaluates how dynamic complementarity across stages of childhood development affects US structural change toward High-Skill Services. It quantifies how the cross-stage amplification embedded in the human capital technology affects structural change by shaping parents’ incentives to invest in their children’s human capital. The key finding is that dynamic complementarity accounts for 64.0% of US structural change toward High-Skill Services in partial equilibrium and 33.1% in general equilibrium. Accordingly, dynamic complementarity embedded in the human capital accumulation technology plays a key role in shaping the quantitative link between early childhood development and structural change.⁷

The fourth counterfactual exercise quantifies the role of human capital inequality in shaping structural change. The headline result is that a mean-preserving 35% increase in human capital dispersion parameters implies just a 20.3 percentage point increase in the High-Skill Services share compared to 25.1 percentage points in the baseline over the same time horizon. A rise in inequality leads to a broad-based decline in parental time devoted to childcare of young children, driving a fall in the supply of skills in the economy.⁸

⁷I supplement both counterfactual exercises with an assessment of their aggregate welfare implications. Across generations and exercises, the distribution-weighted average consumption equivalent variation (CEV) ranges from –66.9% to –6.4% in general equilibrium and from –41.4% to –5.0% in partial equilibrium. Thus, the average welfare loss for parents and their descendants is equivalent to a permanent reduction in baseline consumption of up to 66.9% in general equilibrium and 41.4% in partial equilibrium.

⁸The consumption equivalent variation (CEV) ranges from –16.6% to –9.6%, depending on the generation. Accordingly, the average welfare loss for parents and their descendants is equivalent to a permanent

Related Literature. This paper contributes to three strands of literature. The first relates early childhood human capital to macroeconomic outcomes (Yum, 2023; Daruich, 2018; Lee and Seshadri, 2019; Manuelli and Seshadri, 2014; Schoellman, 2016; Caucutt and Lochner, 2020; Youderian, 2019; Croix and Licandro, 2013). This is motivated by growing evidence that early childhood development is influential in driving life-cycle outcomes (Deming, 2009; García et al., 2017; Cunha and Heckman, 2007; Cunha et al., 2010). Of these papers, the one closest to mine is Schoellman (2016), which uses variation in the ages at which Indochinese migrants arrived in the US to study the effects of an additional year of early childhood in a rich economy setting. The findings suggest that early childhood development is not a strong determinant of cross-country income differences. My paper is complementary in the sense that rather than the country environment, I focus explicitly on the importance of parental time inputs for early childhood human capital and, ultimately, structural change. This is justified by the growing body of evidence that stresses the importance of parental childcare time for early childhood human capital outcomes (Francesconi and Heckman, 2016; Del Boca et al., 2014; Bono et al., 2016; Guryan et al., 2008; Fiorini and Keane, 2014; Bernal, 2008).⁹

The second strand of literature is on the sources of structural change. The two traditional theories of structural change stress the role of income effects (Kongsamut et al., 2001) and unbalanced sector-level growth (Ngai and Pissarides, 2007). In this paper, I focus on the role of human capital in structural change, and therefore contribute to a growing literature on supply-side drivers of structural change (Caselli and Coleman II, 2001; Porzio et al., 2022; Herrendorf and Schoellman, 2018; Acemoglu and Guerrieri, 2008). In this respect, the closest paper to mine is Porzio et al. (2022). They model structural change in the context of an overlapping generations structure, whereby workers in different cohorts self-select into either agriculture or non-agriculture. Human capital in the model is drawn from a distribution that is a function of a cohort-specific component that evolves with the return to skills as well as an individual component that is exogenously determined. In contrast, I endogenize the human capital distribution through parental time investments, allowing my model to fully capture the two-way feedback between structural change and human capital. Furthermore, I enrich my framework with a multi-stage law of motion

reduction in baseline consumption of up to 16.6%.

⁹Parental time allocation is known to shape the human capital formation of children, especially in early stages of childhood. This link operates through several channels. One such channel is through language acquisition and cognitive stimulation, as evidenced by Hart et al. (1997) and Villena-Roldan and Ríos-Aguilar (2012). Another such channel is through problem-solving abilities; through prompts and structure, parents are able to calibrate support in real time to their children such that they can solve problems beyond their unassisted capacity. See Vygotsky (1978) for a theoretical discussion and Pratt et al. (1988) and Neitzel and Stright (2003) for empirical evidence. A final channel operates through stress regulation. Blair and Raver (2012) and Belsky and Fearon (2002) both discuss the role of sustained sensitive caregiving and attachment security for the development of executive functions and cognitive competence of children.

for human capital. That way, the model is well-equipped to study the effects of early childhood development on structural change.

Finally, the paper also contributes to the literature linking structural change with inequality (Comin et al., 2020; Bárány and Siegel, 2018; Buera et al., 2022; Lee and Shin, 2017; Wiedemann, 2026). The closest paper to mine is Wiedemann (2026) because we both show quantitative evidence that the link between inequality and structural change is bidirectional. Furthermore, both papers rely on non-homothetic preferences as one explanation for the link between inequality and structural change (labeled there as the luxury/necessity channel). However, both papers differ in important ways. First, I focus on the role of human capital inequality as opposed to income inequality. Second, the supply-side mechanism that shapes the link between inequality and structural change stems from dynamic complementarity embedded in the human capital technology. In Wiedemann (2026), the supply-side mechanism is driven by the marketization of Services which results from changes in relative prices as inequality rises.

Outline. The rest of the paper is organized as follows. Section 2 presents the empirical facts that motivate the theory. Section 3 presents the full general equilibrium model with heterogeneous agents, multiple sectors and endogenous human capital accumulation. Section 4 calibrates the model while Section 5 uses the quantitative version of the model to perform the counterfactual exercises. Section 6 offers concluding remarks and policy implications.

2 Empirics

This section documents novel cross-country evidence on differences in parental time investments in young children (ages 0–4) and early childhood development outcomes to motivate the general equilibrium model in the subsequent section.

2.1 Data Sources

This section compiles data from four sources. The first data source is the latest version (v11) of the Multinational Time Use Survey (MTUS) (Lamote de Grignon Perez et al., 2024). The MTUS is a harmonized time use survey that contains records on how diarists spent 1440 minutes on a given single day. The MTUS has two types of data. The first is the harmonized aggregated file which contains day-level data (i.e. how much diarists spent on each activity in a single day) while the second is the harmonized episode file data which contains episode-level data (i.e. an episode is a change in activity, location, transport,

co-presence or the use of technological devices). I rely on the harmonized aggregated file given that I am interested in better understanding aggregate time use trends. The time use data is accompanied by a number of useful variables including the age, sex and education level of the diarist as well as data on the number of household members, the number of children below the age of 18 living the household and the age of the youngest child living in the household. Accordingly, each observation in the harmonized aggregated file data corresponds to a country-year-person-day observation. Hereafter, I refer to ‘MTUS’ and ‘time use’ data interchangeably.

The second data source is the United Nations International Children’s Emergency Fund’s (UNICEF) Multiple Indicator Cluster Survey program accessed through the Integrated Public Use Microdata Series (IPUMS-MICS) (Bolgrien et al., 2024). IPUMS-MICS offers harmonized cross-country survey-level data on children worldwide. The dataset focuses on a diverse set of topics, including early childhood development. The IPUMS-MICS dataset is split into multiple units of analysis including births, children between the ages of 0 and 4 years old, children between the ages of 5 and 17 years old and women. The early childhood development module contains data on parental time investments in children and early childhood human capital outcomes.

The third dataset is the IPUMS International (IPUMS-I) dataset (Ruggles et al., 2024). This dataset contains rich household- and person-level data that is harmonized across country-year samples. The list of variables recorded is large, ranging from nativity and birthplace to occupation and industry of work. I use the person-level census data with an explicit focus on the occupation and industry modules for information on the individual’s primary occupation and industry of work.

The fourth dataset is the Penn World Tables v11.0 (Feenstra et al., 2015) which I use to obtain data on PPP-adjusted real GDP per capita at the country-year level — a standard proxy for development.

2.2 Cross-Country Patterns

To motivate my theory, I start by showing that the process of economic development is accompanied by both a rise in the supply of skilled labor and the employment share in High-Skill Services. The x-axis in Figure 1 corresponds to the log of PPP-adjusted real GDP per capita while the y-axis corresponds to the population-weighted share of total employment of skilled workers. Skilled workers are defined, following Acemoglu and Autor (2011), as workers who belong to managerial, professional or technical occupations.¹⁰ In contrast to educational attainment — a common proxy for a worker’s skill level — the

¹⁰These are identified as observations whose primary occupation is either legislator/senior official/manager, professional or technician/associate professional which is identified by the OCCISCO variable.

occupation captures the multi-dimensional nature of skill content, including educational attainment and wage levels. Figure 1 shows that along the development path, the share of skilled workers rises. This is confirmed by a simple regression of the skilled workers share against the log of real GDP per capita. The regression reveals that a 1% increase in real GDP is associated with a 0.088 percentage point in the share of skilled workers in the economy. This fact has been documented before; Katz and Murphy (1992), Acemoglu and Autor (2011) and Buera et al. (2022) are among the papers that have done so in part to understand why the wage premium rises with the relative supply of skilled labor. Figure 1 is instead helpful in showing that this phenomenon holds for a broad set of countries.

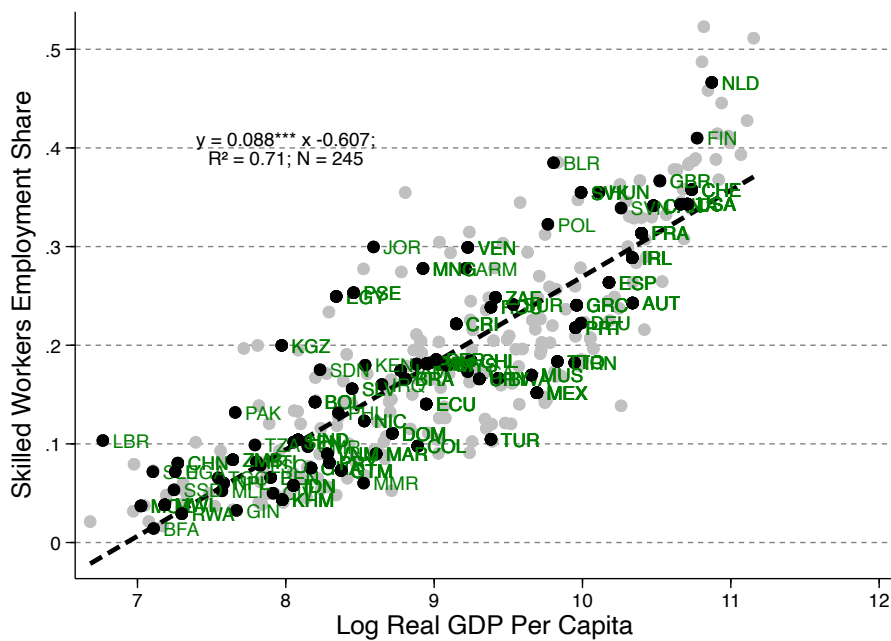


Figure 1: Skilled Workers Employment Share

Notes: Grey circles are country-year observations. Black circles are country averages. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the share of skilled workers on log PPP-adjusted real GDP per capita using the country-year observations. *(**)[***] statistically significant at the 10% (5%) [1%] level.

I then split Services industries in the INDGEN variable into High- and Low-Skill Services sectors based on the industry's share of total employment coming from skilled occupations, focusing only on the US censuses as the benchmark. Table A.1 in the Appendix presents the results of the industry classification by the 1960-2020 average skilled share of employment. There is a clear gap in the average skilled share of employment between the first four and the latter six Services industries, generating a natural threshold that places Education, Health and Social Work, Financial Services and Insurance and

Business Services and Real Estate in the High-Skill Services bin. This classification appears consistent with every individual census year in addition to the 1960-2020 average.

Figure 2 confirms the unambiguous positive correlation between the High-Skill Services share of employment and development documented by Buera and Kaboski (2012) and Buera et al. (2022). A simple regression of the High-Skill Services share of employment on the log of PPP-adjusted real GDP per capita shows that, on average, a 1% increase in the latter is associated with a rise in the former by 0.075 percentage points. This is not surprising given the current state of knowledge on the process of structural transformation; the early works of Kuznets (1957), Fuchs (1968) and Baumol (1967) among others showed a shift in economic activity towards Services over the development path. More recent papers such as Buera and Kaboski (2012), Fang and Herrendorf (2021) and Buera et al. (2022) have focused exclusively on the sources and consequences of the growing importance of High-Skill Services with development, and so this is not the first paper to document the growing importance of High-Skill Services accompanying the process of growth and development. Like Figure 1 however, Figure 2 is helpful for showing that this phenomenon holds for a broad panel of countries across the development spectrum.

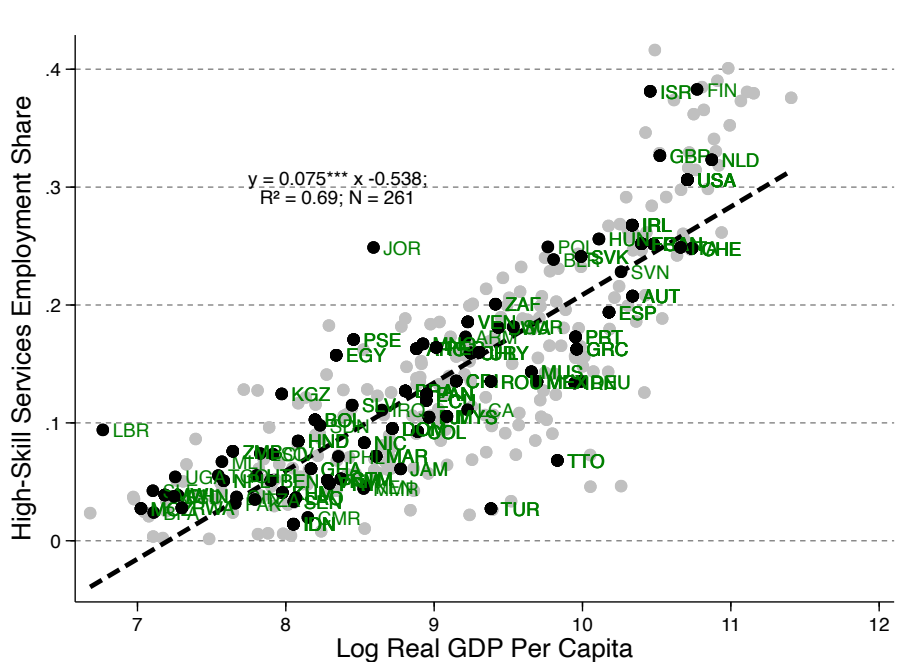


Figure 2: High-Skill Services Employment Share

Notes: Grey circles are country-year observations. Black circles are country averages. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the share of High-Skill Services on log PPP-adjusted real GDP per capita using the country-year observations. *(**)[***] statistically significant at the 10% (5%) [1%] level.

Next, I document a novel fact: the parental time investments in children below the age of 5 rise with development. Childcare in the time use survey is defined as physical and/or medical childcare, teach and/or help with homework, read, talk and/or play with child, and supervise, accompany and/or other childcare. I restrict my attention to individuals who are between 18 and 64 years of age (i.e. working-age individuals) and who live in a household with a child between the ages of 0 and 4 years old (inclusive).¹¹ Figure 3 plots the country-year population-weighted average time spent on early childhood childcare against log PPP-adjusted real GDP per capita and shows that richer countries, on average, devote more time to childcare. To gauge the magnitude of these cross-country income differences, Americans earned 4 times as much as South Africans on average in 2010, and spent almost three times as much time on childcare in the same year (close to 2 hours compared to 42 minutes). Indeed, the regression shows that the semi-elasticity of 0.020 is statistically significant at the 1% level.

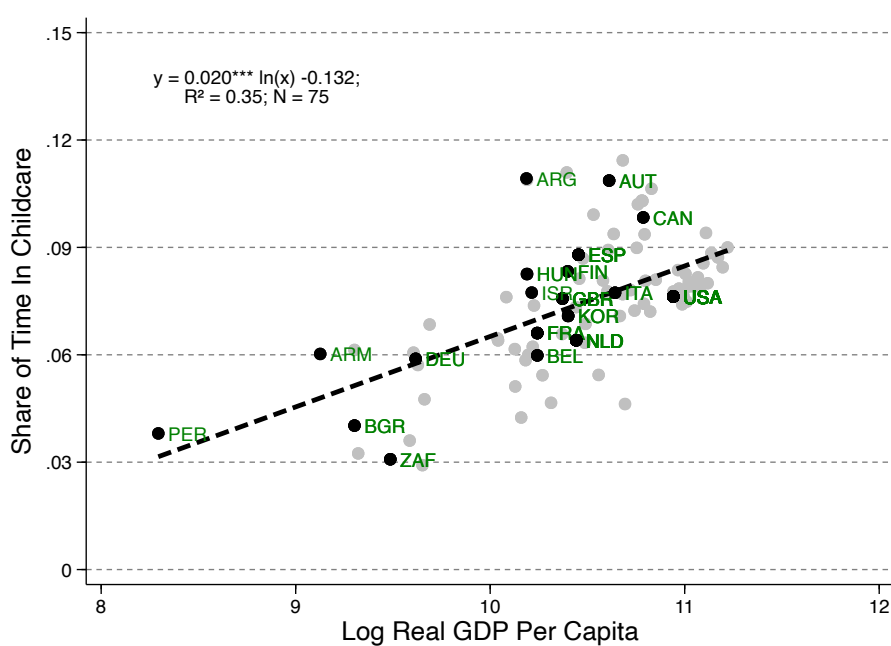


Figure 3: Share of Time Spent on Childcare

Notes: The underlying data restricts attention to individuals aged between 18 and 60 (inclusive). Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights to correct for differences in days of the week. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the share of time spent on childcare on log PPP-adjusted real GDP per capita using the country-year observations. *(**)[***] statistically significant at the 10% (5%) [1%] level.

¹¹It is not possible to identify the diarist's relationship with the child (i.e. whether the diarist is the mother, father, sibling, etc. of the child) The only observables are the age of the youngest child living in the household and the number of children (below 19 years of age) in the household.

Figure 3 is distinct from other attempts to document cross-country patterns of parental time investment in childcare in important ways. For instance, I find that there is a clear positive link between time devoted to childcare and development. This differs from [Bick et al. \(2018\)](#) who find that this relationship is not monotonic. Relative to [Bick et al. \(2018\)](#), I add a restriction on the diarists to ensure that a child lives in the same household. That way, I can abstract from fertility trends and, by extension, capture the intensive margin of parental time investments. While my result is similar to that of [Guryan et al. \(2008\)](#), my dataset differs in that I keep diarists who live with a child less than 5 years of age while they keep diarists who live with a child less than 18 years of age. This distinction results from my interest in studying cross-country differences in early childhood development patterns, where parental time is a key input. Second, I include a greater number of countries in my sample.

Figure 3 can benefit from a number of robustness checks. To fully abstract from fertility trends and focus on the intensive margin, I also plot the same variable with an additional restriction that the number of children in the household is 1. Appendix A.2 shows that, qualitatively, the same trend holds and that, quantitatively, the regression coefficient is very close to that in Figure 3, ruling out any conclusion relating this novel fact with fertility trends. Another potential challenge to Figure 3 is that while offering an expanded set of countries relative to previous attempts, it is not clear if this fact is robust across the entire development spectrum. To tackle this challenge, I present data from the IPUMS-MICS. This dataset asks mothers and fathers of children aged between 0 and 4 years of age six binary questions each on whether or not the parent in question has undertaken a given time-intensive activity with that child in the last 3 days. After aggregating the binary responses for both parents at the country-year level using children under 5 sample weights, I then plot these against log PPP-adjusted real GDP per capita.¹² The results are presented in Appendix A.3 and confirm the empirical fact that richer countries dedicate more time to early childhood childcare, even on the lower end of the development spectrum. It is also helpful to confirm that these results hold within-country. While it is a well-established fact in the literature that time allocated to childcare rises over time in the US ([Aguilar and Hurst, 2007](#); [Guryan et al., 2008](#); [Ramey and Ramey, 2009](#)), I present the plot in Appendix A.4 using the same data as Figure 3 (focusing on children between the ages of 0 and 4 inclusive) for the sake of completeness. Barring 1975, there is a consistent increase in time spent on childcare over time implying that this phenomenon is observed both within and across countries over the development process.

Figure 4 plots the share of time spent on childcare against log PPP-adjusted real GDP

¹²The IPUMS-MICS dataset does not offer any data on minutes spent on childcare by parents, but the literature on early childhood development notes that parental time investments can be (and are frequently) measured by the frequency of activities (see [Attanasio et al., 2022](#)).

per capita by gender. It shows that on average both men and women spend more time with young children as the economy develops. The correlation between both variables is statistically significant at the 1% level for both genders indicating that this is not a gender-specific phenomenon. Appendix A.5 verifies that these gender-specific trends also arise from the IPUMS-MICS dataset.

The difference in levels between men and women is not surprising given that these trends coincide with a number of important gender-biased structural changes accompanying the development process including changes in female labor force participation (Goldin, 1994) and home production (Bridgman et al., 2018). However, that the time allocated to the childcare of young children rises between females suggests that this pattern could be separated from female labor force participation trends given that greater participation should intuitively induce a decline in time allocated to the childcare of young children. Appendix A.6 shows that the childcare time allocation share of females increase with development regardless of employment status, suggesting that this fact is robust to female labor force participation trends.

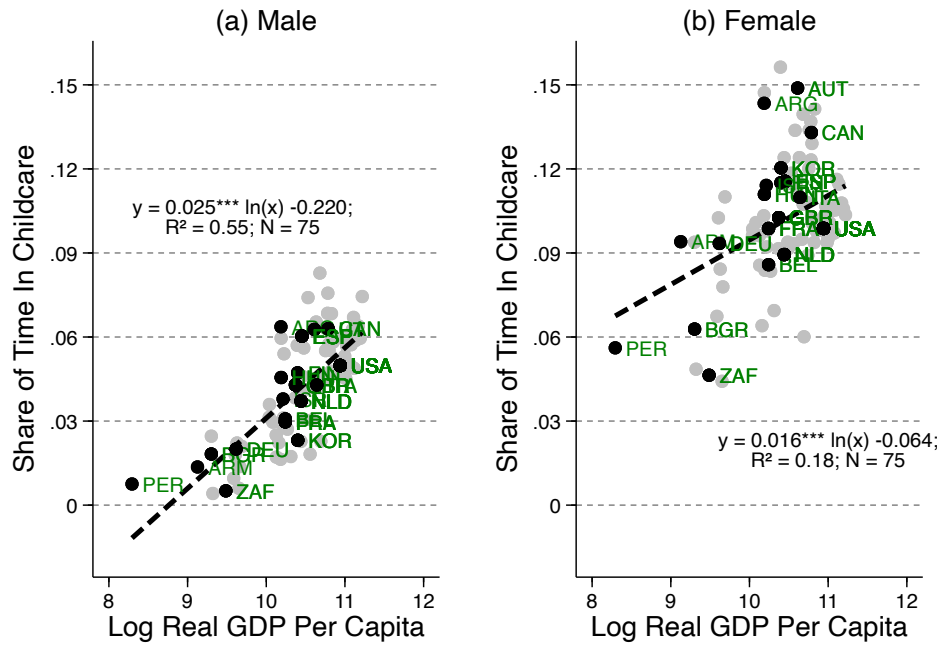


Figure 4: Share of Time Spent on Childcare By Gender

Notes: Panel (a) plots the share of time allocated to childcare towards young children (ages 0–4) by males. Panel (b) plots the share of time allocated to childcare towards young children (ages 0–4) by females. The underlying data restricts attention to individuals aged between 18 and 60 (inclusive). Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights to correct for differences in days of the week. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses for each gender the share of time spent on childcare on log PPP-adjusted real GDP per capita using the country-year observations. ** *** statistically significant at the 10% (5%) [1%] level.

I then document another novel fact: early childhood development outcomes vary with development. To the best of my knowledge, this is the first attempt to do so at the cross-country level. To measure these outcomes, I use the IPUMS-MICS dataset which asks respondents three binary questions: (i) can the child read at least four simple, popular words? (ii) can the child identify at least 10 letters of the alphabet? and (iii) can the child name and recognize symbols of all numbers from 1-10? Aggregating the number of yes values at the child-level (assigning equal weights to each question) and then aggregating again at the country-year level using the proposed under-5 survey weights yields a value from 0 to 3 for each country-year observation. I plot these against log PPP-adjusted real GDP per capita in Figure 5. Since the speed of cognitive development can be closely related to age when kids are youngest, I separate children that are 3 years of age and children that are 4 years of age.

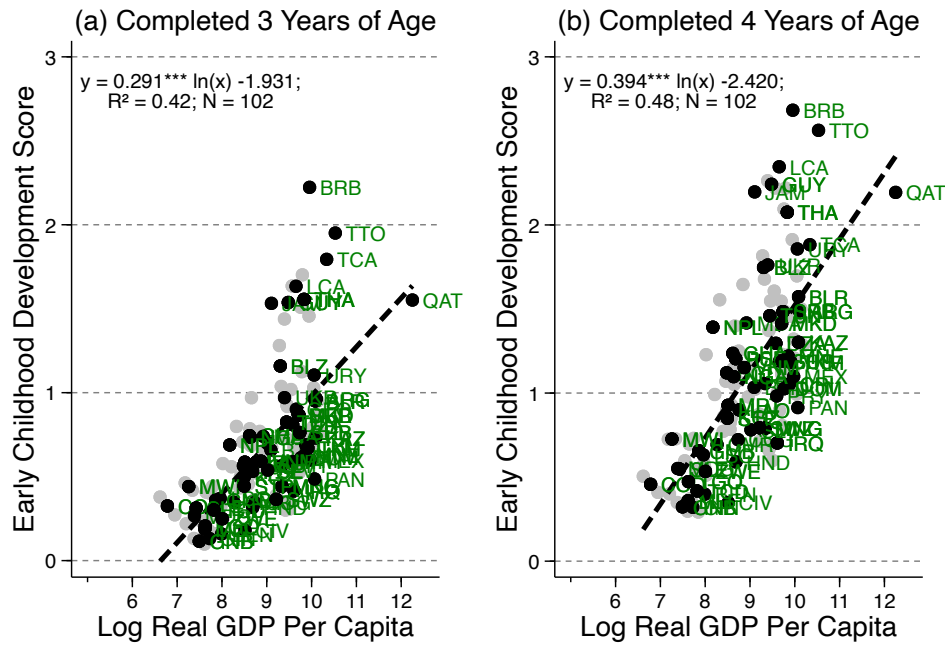


Figure 5: Early Childhood Development Outcomes

Notes: Panel (a) plots the early childhood development score of children that have completed three years of age. Panel (b) plots the early childhood development score of children that have completed four years of age. Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights to correct for differences in days of the week. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses for each educational group the share of time spent on childcare on log PPP-adjusted real GDP per capita using the country-year observations. ** *** statistically significant at the 10% (5%) [1%] level.

The correlation between both variables is visibly strong both qualitatively and quantitatively. As expected the link is stronger for 4 year-old respondents. On average, a Thai 4 year-old will be able to perform two of the three tasks correctly while their Zimbabwean counterpart is more likely than not to be unable to perform any of these tasks correctly. This example not only highlights that cross-country differences exist in early childhood development outcomes, but that these differences are quite significant even when comparing two countries at the lower end of the development spectrum.

2.3 Summary

This section motivates the interest in High-Skill Services by showing that the High-Skill Services share of employment rises considerably over the development process, eventually taking up almost half of total employment in the richest countries. This coincides with the rise in the supply of skills in the economy along the development path. This section

then documents novel empirical facts related to cross-country parental time investments in young children and early childhood development outcomes. The two novel facts are that parents in richer countries on average dedicate more time to their young children and these young children have better early childhood development outcomes.

3 Model

To uncover and quantify the causal link between early childhood development and structural change, I construct a heterogeneous agents general equilibrium framework. The modeling exercise combines an intertemporal multi-stage human capital accumulation problem into a benchmark multi-sector model of growth and structural change in order to jointly account for time use, early childhood human capital, and structural change patterns.

3.1 Environment

Assume an infinite time horizon where time is denoted by $t \in \mathbb{Z}_{\geq 0}$. At any given time t , there are three sectors that operate in this economy: Goods (G), Low-Skill Services (L) and High-Skill Services (H). Each sector has a representative firm that operates in a perfectly competitive environment, and each firm decides how many efficiency units of skilled (S) and unskilled (N) labor to hire in their respective static profit-maximization problems.

Let the economy be populated by a fixed continuum of dynasties i of measure $\mathcal{L}$ and let $g \in \{1, \dots, \infty\}$ serve as a generation superscript. At any given time t , a dynasty consists of a single parent and a single child (also referred to as a household hereafter).¹³ The parent is the only decision-maker inside the household. Children do not work nor consume, parents invest time to develop their child's human capital because they derive direct utility (i.e. enjoyment) from it and because they attach value to their child's future outcomes. Parental time investments are the only human capital investments allowed in this economy, and the elasticity of this input varies with the child's stage of human capital development.

Time allocated to childcare has an opportunity cost of time at work which allows parents to earn an income which they can spend on the consumption of different sectoral outputs.¹⁴ A parent working as a skilled worker contributes their human capital level as efficiency units of labor per unit of time while all parents supply a single efficiency unit of labor per unit of time if they work as an unskilled worker regardless of their human capital

¹³This choice of demographics is motivated by the empirical facts; the rise in the share of time parents spend with their children occurs on the intensive margin (i.e. per child) and is observed for both mothers and fathers.

¹⁴I therefore use the terms “worker” and “parent” interchangeably.

level and the sector they are employed in. At any given time t , parents belong strictly to one sector and occupation. Accordingly, at all times the incumbent parent of household i seeks to maximize utility by solving the following three problems: (i) a decision on how much time to invest in their child; (ii) a decision on how much to consume from each sector; and (iii) a decision on which occupation $\{S, N\}$ and sector $\{G, L, H\}$ to work in.

All households live hand-to-mouth in this economy as there are no financial markets. There is free labor mobility across both sectors and occupations for workers, and all markets clear at all times.

Timing and Heterogeneity. Each individual lives for $2T$ years. For the first T years, the individual is a child. For the subsequent T years, the individual is a parent to a child of their own.¹⁵ The cycle repeats perpetually for each dynasty over the infinite time horizon. At any time t , an individual belonging to a given generation g is $t - [(g - 1) \times T]$ years old. Let $\mu(h, h'; t)$ denote the joint probability density function at time t of a generic parent-child human capital pair $h_i^g = h$ and $h_i^{g+1}(t) = h'$.

Upon birth, the child of dynasty i draws an initial human capital level $h_i^{g+1}(t = gT)$ from a lognormal distribution $G(\eta_c, \kappa_c)$ where η_c controls the location and κ_c controls the scale of the distribution. This distribution is time- and dynasty-invariant, and is known to all parents. Furthermore, each parent observes their child's draw upon birth. The human capital of the child from birth to age T evolves through the multi-stage law of motion for human capital accumulation, after which time the human capital level is fixed for the remainder of the individual's life. Consequently, at any time t , households are ex-ante heterogeneous in parent-child human capital pairs $h_i^g, h_i^{g+1}(t) \in \mathcal{H} \times \mathcal{H}$.

Households. At every time t , each parent with parent-child human capital pair $h_i^g, h_i^{g+1}(gT)$ is endowed with a single unit of time to be allocated between two activities: working $n_i^g(t)$ and spending time with the child $\tau_i^g(t)$. Parents seek to maximize the present discounted utility at all times. For a given dynasty, I specify the following discrete-time life-cycle structure in recursive formulation for preferences from the point of view of a parent of household i belonging to generation g

¹⁵The life-cycle notation and structure here is similar to (Cunha and Heckman, 2007).

$$\begin{aligned}
V_i^g(h_i^g, h_i^{g+1}(gT)) = \max_{\{\tau_i^g(t)\}_{t=gT}^{(g+1)T-1}} & \left\{ \underbrace{\sum_{t=gT}^{(g+1)T-1} \beta^{t-gT} U(c_i^g(t), \tau_i^g(t))}_{\text{current generation utility}} \right. \\
& \left. + \underbrace{\beta^T \beta_g \mathbb{E}_{h_i^{g+2}} \left[V_i^{g+1}(h_i^{g+1}((g+1)T), h_i^{g+2}((g+1)T)) \right]}_{\text{future generations' utility}} \right\}, \tag{1}
\end{aligned}$$

where β is the time discount factor, β_g is the inter-generational discount factor and $c_i^g(t)$ is the household's aggregate consumption. It is easy to see that the current formulation for preferences suggests that the parent internalizes the utility of future generations of the dynasty: the first term is the present discounted value of utility of the parent belonging to generation g while the second term is the continuation value which collects the sum of lifetime utility of *all* descendants discounted by an additional inter-generational discount factor β_g . The parent of generation g only observes the human capital draw of their child of generation $g+1$ and not that of the generation $g+2$ child. Accordingly, the parent takes expectations over the latter using the time-invariant initial human capital distribution $G(\eta_c, \kappa_c)$ known to the current parent.

The utility function is concave, strictly increasing and twice differential with $U : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. I assume the familiar [MaCurdy \(1981\)](#) structure for instantaneous utility¹⁶

$$U(c_i^g(t), \tau_i^g(t)) = \frac{c_i^g(t)^{1-\nu}}{1-\nu} - \omega \frac{n_i^g(t)^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}}. \tag{2}$$

The parameter ω is the time-invariant weight that the parent attaches to the pure dis-satisfaction generated from work. The choice to model a disutility for hours worked (i.e. utility from time spent with children, akin to leisure) in the first place is motivated by the findings of [Cosaert and Hennebel \(2023\)](#) who, using Dutch consumption and time use data, find that 40% of parental childcare time allocation is perceived as a form of leisure. ν is the usual CRRA parameter which controls the curvature in consumption. It can be shown that this parameter will determine the strength of income effects; if $\nu > 1$, then a rise in a given parent's income will prompt a decline in hours worked and vice versa.

Aside from their time allocation, the parent also chooses the sectoral composition of

¹⁶This utility function choice is consistent with the wider literature on time allocation and aggregate labor supply over time (see [Bick et al. \(2022\)](#) and [Boppart et al. \(2023\)](#) for some examples of this). [Boppart and Krusell \(2020\)](#) show that [MaCurdy \(1981\)](#) preferences can generate a balanced growth path with falling hours worked with income.

aggregate consumption. Letting $c_i^g(t)$ be a composite of consumption from three sectors: Goods ($c_{i,G}^g(t)$), Low-Skill Services ($c_{i,L}^g(t)$) and High-Skill Services ($c_{i,H}^g(t)$), I assume the following implicitly defined CES aggregator from [Comin et al. \(2021\)](#)

$$\sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad (3)$$

where θ_j is a time-invariant sector-specific preference weight, $c_{i,j}^g(t)$ is the sector-specific quantity consumed and $\sigma \in (0, \infty)$ is the (constant) elasticity of substitution that governs the degree of substitutability (if $\sigma > 1$) or complementarity (if $\sigma < 1$) between sectoral consumption. Crucially, λ_j influences the relative income elasticities of sectoral demand. Consequently, this will govern the extent to which the sectoral composition of the consumption basket will vary with household income. The advantage of (3) relative to alternative consumption aggregators such as Stone-Geary preferences is that, as will become clearer when studying the equilibrium path of this economy, income effects do not level off but rather persist in the long-term, better capturing long-run Engel curves.

Finally, each parent self-selects into sector $\{G, L, H\}$ and occupation of work $\{S, N\}$ at any given time t . Parents who choose a skilled occupation contribute h_i^g units of efficiency labor while those that choose an unskilled occupation contribute exactly 1 unit of efficiency labor regardless of their human capital level. Accordingly, the parent's gross income at time t , $y_i^g(t)$, belongs in the following set

$$y_i^g(t) \in \{w_{j,S}(t)h_i^g, w_{j,N}(t) : j \in \{G, L, H\}\}, \quad (4)$$

where $w_{j,S}(t)$ and $w_{j,N}(t)$ are the wage for a single unit of efficiency labor as a skilled and unskilled worker in sector j , respectively. Since households live hand-to-mouth in this economy, the following budget constraint must hold at all times

$$(1 - \tau_i^g(t))y_i^g(t) \geq p_i^g(t)c_i^g(t), \quad (5)$$

where $p_i^g(t)$ is the price of aggregate consumption.

Human Capital Accumulation. Inspired by [Del Boca et al. \(2014\)](#) and [Cunha et al. \(2010\)](#), parents have access to a multi-stage human capital production function for their children. The child's initial human capital is drawn from a time- and dynasty-invariant lognormal distribution $G(\eta_c, \kappa_c)$ where η_c and κ_c control the location and scale of the distribution, respectively. Each parent knows the distribution and observes the child's

draw at all times.¹⁷ Let $s \in \{1, \dots, S\}$ denote the stage of childhood development that a child is currently in. The child's human capital evolves according to

$$h_i^{g+1}(t+1) - h_i^{g+1}(t) = h_i^{g+1}(t)^{\zeta_s + q} \left(h_i^g \tau_i^g(t) \right)^{1 - \zeta_s}, \quad t \in \{gT, \dots, (g+1)T - 1\}, s = t - gT + 1 \in \{1, \dots, S\}. \quad (6)$$

The first input $h_i^g \tau_i^g(t)$ is the parent's effective time allocation to childcare. This is composed of the units of time that the parent of household i and generation g chooses to spend with their child, $\tau_i^g(t)$, and the parent's fixed human capital stock h_i^g which acts as a productivity term to the childcare time allocation. The second input is the current human capital level of the child, $h_i^{g+1}(t)$. These inputs typically feature in multi-stage human capital models in a Cobb-Douglas fashion, as in [Del Boca et al. \(2014\)](#). Crucially, the relative intensities of these inputs (governed by $1 > \zeta_s > 0$) vary with the stage of childhood development, s , to capture the varying levels of importance associated with each input over the childhood development period. Relative to [Del Boca et al. \(2014\)](#), I add a human capital persistence term q . This regulates the degree of both self-productivity and dynamic complementarity and therefore allows for a more flexible parameterization to fit the data.¹⁸ As soon as the child becomes a parent at the age of T , the human capital level is fixed for the remainder of life. For simplicity, there is no depreciation in human capital in this economy.

Production. On the production side¹⁹, there is a profit-maximizing sector-specific representative firm that operates in a perfectly competitive environment. The firm produces sectoral output $Y_j(t)$ by augmenting aggregate efficiency units of labor $Z_j(t)$ using a sector-specific technology $A_j(t)$ which grows at a constant growth rate g_j known to all parents at all times

$$Y_j(t) = A_j(t)Z_j(t) \quad \forall j \in \{G, L, H\}. \quad (7)$$

¹⁷An alternative modeling choice would be to assume that each parent does not observe the child's draw and instead makes decisions based on the expectation of the distribution such that the child's human capital is realized ex-post all investment decisions (i.e. when they become parents themselves). To remain consistent with the literature on childhood development (e.g. [Del Boca et al., 2014](#)), I assume that the parent observes the child's draw.

¹⁸Mathematically, dynamic complementarity requires that the cross-partial derivative of the human capital production function be positive (i.e. $\frac{\partial^2 h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t) \partial \tau_i^g(t)} > 0$) while self-productivity requires that the derivative of the human capital production function with respect to human capital be positive (i.e. $\frac{\partial h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t)} > 0$). This setup implies that children with better early childhood human capital are more likely to draw in greater investments by their parents in later stages of their childhood, fixing the parent's human capital level.

¹⁹Since only the parent can work in this economy, the generation indices are dropped hereafter.

Sectors differ by their skill intensity: (i) Sector L hires both skilled and unskilled labor. Both types of labor are aggregated using a CES technology with an elasticity of substitution $\epsilon \in (0, \infty)$ which determines the extent to which both types of labor are substitutes (if $\epsilon > 1$) or complements (if $\epsilon < 1$) and time-invariant technology weights $0 < \phi_L < 1$ which influence the skill intensity of the sector

$$Z_L(t) = \left[\phi_L N_L(t)^{\frac{\epsilon-1}{\epsilon}} + (1 - \phi_L) S_L(t)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \quad (8)$$

where $S_L(t)$ and $N_L(t)$ are the aggregate efficiency units of skilled and unskilled labor.

(ii) Sector G has a technology weight of $\phi_G = 1$ and hence only hires unskilled labor

$$Z_G(t) = N_G(t). \quad (9)$$

(iii) Sector H has a technology weight of $\phi_H = 0$ and therefore only hires skilled labor

$$Z_H(t) = S_H(t). \quad (10)$$

This set-up implies that sector H is the most skilled labor-intensive sector while sector G is the most unskilled labor-intensive sector. Output from sector $j \in \{G, H, L\}$ is sold to households at a price $p_j(t)$.

3.2 Equilibrium

Sectoral Producers. The three sectoral producers choose the quantities of skilled, $S_L(t)$, and unskilled labor, $N_L(t)$, to maximize profits given their technologies, output prices and wages. The H -sector producer only hires skilled labor, and so the corresponding optimality condition reveals that the producer will hire skilled labor up until the marginal value product of labor is equated with the wage rate

$$p_H(t)A_H(t) = w_{H,S}(t). \quad (11)$$

Likewise, the G -sector producer only hires unskilled labor, and so demand for labor is also perfectly elastic with

$$p_G(t)A_G(t) = w_{G,N}(t). \quad (12)$$

The L -sector representative producer on the other hand hires both types of labor. Taking first order conditions of the L -sector producer's profit maximization problem reveals that the relative demand for skilled labor in that sector depends positively on the relative skill intensity of that sector and the relative wage of skilled labor in equilibrium

$$\frac{N_L(t)}{S_L(t)} = \left(\frac{(1 - \phi_L) w_{L,S}(t)}{\phi_L w_{L,N}(t)} \right)^\epsilon. \quad (13)$$

As usual, this relationship is regulated by the elasticity of substitution between both types of labor.²⁰

Parents. Parents decide (i) which occupation and sector to work in, (ii) how much time to invest in their child and (iii) how much of each sector to consume given preferences, wages and prices. Proposition 1 simplifies the household's problem by proving that a utility-maximizing parent is also a gross income-maximizing parent.

Proposition 1 (Separation of occupational choice) *Suppose that the occupational choice problem only affects the parent's problem through gross income. Then, the parent's problem can be solved sequentially. First, the parent makes the occupational choice that maximizes gross income. Second, the parent jointly chooses time allocation and sectoral quantities that maximize preferences (1) subject to (5) and $n_i^g(t) + \tau_i^g(t) \leq 1$.*

Proof. Any feasible allocation of time and sectoral quantity under gross income $y(t)$ remains feasible under a gross income $y'(t) > y(t)$ since $y'(t)$ expands the parent's budget without affecting preferences or the time constraint. See Appendix B.2 for formal proof. $\square$

Proposition 1 allows (4) to be written as a standard Roy selection-type problem

$$y_i^g(t) = \max \{ w_{j,S}(t) h_i^g, w_{j,N}(t) : j \in \{G, L, H\} \}, \quad (14)$$

where given wages and the fixed human capital level, parents choose the sector and occupation that maximizes their gross income. With gross income determined, the parent's problem boils down to a standard two-part structural change model where $\forall t$ the given parent chooses $\tau_i^g(t)$ to maximize intertemporal preferences (1) subject to the budget constraint (5) (i.e. the intertemporal problem) and then chooses the sectoral consumption quantities that maximize (3) subject to the budget constraint (5) with net income substituted in (i.e. the static problem). The solution to the static problem yields the standard Marshallian demand functions²¹

$$c_{i,j}^g(t) = \frac{e_{i,j}^g(t) p_i^g(t) c_i^g(t)}{p_j(t)}, \quad \text{where} \quad e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)} \quad \forall j \in \{G, L, H\}. \quad (15)$$

²⁰The proof for (11), (12) and (13) is located in Appendix B.1.

²¹The proof for (15) is located in Appendix B.3.

The demand functions write the sectoral consumption quantity from a given sector as a function of the parent's total expenditure, $p_i^g(t)c_i^g(t)$ (which by virtue of the budget constraint is equivalent to time-adjusted income $(1 - \tau_i^g(t))y_i^g(t)$). Note that the standard income and substitution effects are present in (15). A higher relative price $p_j(t)$ reduces consumption of sector j 's good, while greater aggregate consumption induces a disproportionate change in consumption of sector j 's good by a magnitude controlled by λ_j .²²

Equilibrium. I set the price of the H sector's output $p_H(t)$ as the numéraire in this economy.

Definition 1 (Competitive equilibrium) *Normalize $p_H(t) = 1$ for all $t \in \mathbb{Z}_{\geq 0}$. A competitive equilibrium is a collection of price, wage, and technology sequences*

$$\left\{ \{p_k(t)\}_{k \in \{G,L\}}, \{w_{j,o}(t)\}_{j \in \{G,L,H\}, o \in \{S,N\}}, \{A_j(t)\}_{j \in \{G,L,H\}} \right\}_{t \in \mathbb{Z}_{\geq 0}},$$

together with household allocations and firms' labor demands, such that:

1. *Given prices and wages, the parent in each dynasty i and generation g , with state $(h_i^g, h_i^{g+1}(gT))$, maximizes intertemporal utility by choosing sequences of occupations, sectors of employment, sectoral consumption, and childcare time,*

$$\left\{ o_i^g(t), j_i^g(t), \{c_{i,k}^g(t)\}_{k \in \{G,L,H\}}, \tau_i^g(t) \right\}_{t=gT}^{(g+1)T-1},$$

where $o_i^g(t) \in \{S,N\}$ and $j_i^g(t) \in \{G,L,H\}$, subject to the per-period budget and time constraints.

2. *Given prices, wages, and technologies, the representative producer in each sector $j \in \{G,L,H\}$ maximizes profits in every period $t \in \mathbb{Z}_{\geq 0}$ by choosing its demands for skilled and unskilled labor efficiency units, $S_j(t)$ and $N_j(t)$.*
3. *All sectoral output markets and skilled and unskilled labor markets clear in every period $t \in \mathbb{Z}_{\geq 0}$.*

3.3 Market Clearing

Structural Change. The Roy-type selection problem together with the assumption of free labor mobility across sectors for skilled and unskilled workers implies the continued relocation across sectors until sectoral wages equalize. Since both the G - and L -sector

²²Note that if $\lambda_j = 1$, (15) collapses to a homothetic demand function.

producers hire unskilled labor, unskilled labor wage equalization using these sectors' first order conditions implies

$$\underbrace{p_G(t)A_G(t)}_{w_{G,N}(t)} = \underbrace{\phi_L p_L(t)A_L(t) \left(\frac{Z_L(t)}{N_L(t)} \right)^{\frac{1}{\epsilon}}}_{w_{L,N}(t)} = w_N(t). \quad (16)$$

Wage equalization in the market for skilled labor instead uses the first order conditions from the H - and L -sector producers since they are the only sectors that hire skilled workers. The corresponding wage equalization implies

$$\underbrace{A_H(t)}_{w_{H,S}(t)} = \underbrace{(1 - \phi_L) p_L(t)A_L(t) \left(\frac{Z_L(t)}{S_L(t)} \right)^{\frac{1}{\epsilon}}}_{w_{L,S}(t)} = w_S(t). \quad (17)$$

These two equations ensure that in equilibrium and conditional on choosing an occupation, parents will be indifferent between the sector of work.

Human Capital Allocation. Likewise, the Roy-type selection problem together with the assumption of free labor mobility across occupations implies that a parent with a human capital level h_i^g will pursue skilled work if $h_i^g \geq \bar{h}(t)$ where $\bar{h}(t)$ is constructed using the indifference condition

$$\bar{h}(t) = p_G(t) \frac{A_G(t)}{A_H(t)}. \quad (18)$$

Note that the indifference condition does not depend on time allocation thanks to Proposition 1. The threshold (18) indicates that in addition to the raw human capital level of the parent, the supply of skills in the economy is also pinned down by general equilibrium effects — namely structural change (through relative prices) and sector-biased technical change (given complementarities between labor and TFP in production).

Market Clearing. Recall that $\mu(h, h'; t)$ denotes the joint probability density function at time t of a generic parent-child human capital pair $h_i^g = h$ and $h_i^{g+1}(t) = h'$ and that $\mathcal{L}$ represents the (fixed) mass of dynasties in this economy. The sectoral goods market clearing conditions can be written as

$$A_j(t)Z_j(t) = \mathcal{L} \int_h \int_{h'} c_j(h, h'; t) \mu(h, h'; t) dh' dh \quad \forall j \in \{G, L, H\}, \quad (19)$$

where the LHS is the supply and the RHS is the demand for a given sector's output.

Meanwhile, the market clearing conditions for skilled and unskilled labor can be written as

$$\sum_{j \in \{G, L, H\}} S_j^d(t) = \mathcal{L} \int_{\bar{h}(t)}^{\infty} \int_{h'} h(1 - \tau(h, h'; t)) \mu(h, h'; t) dh' dh \quad \text{and} \quad (20)$$

$$\sum_{j \in \{G, L, H\}} N_j^d(t) = \mathcal{L} \int_0^{\bar{h}(t)} \int_{h'} (1 - \tau(h, h'; t)) \mu(h, h'; t) dh' dh, \quad (21)$$

where the LHS is the sum of skilled and unskilled efficiency units of labor demanded by the three firms while the RHS is the time-adjusted supply of skilled and unskilled efficiency units of labor, respectively.

3.4 Steady State

Sections 3.2 and 3.3 characterize the equilibrium conditions that must hold in the transitional dynamics of this economy. Motivated by the growing importance of High-Skill Services with development, the remainder of this section studies the properties of an economy where all activity is occupied by the H -sector.

Proposition 2 (Asymptotic Dominance of the H -sector) *Suppose that $\lambda_H > \lambda_L, \lambda_G, g_L, g_G > g_H > 0$ and that $0 < \sigma < 1$. For $k \in \{L, G\}$ define*

$$x_k \equiv -\sup_{t \geq 0} \left[(1 - \sigma)(g_H - g_k) + (1 - \sigma)(\lambda_k - \lambda_H) \frac{\dot{c}_i^g(t)}{c_i^g(t)} \right] > 0.$$

- *If $\epsilon = \infty$: the parameter restrictions are sufficient for asymptotic dominance of the H -sector.*
- *If $\epsilon < \infty$: the H -sector attains asymptotic dominance provided that, in addition, for each $k \in \{L, G\}$,*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t \frac{(1 - \sigma)}{\epsilon} \left(\frac{\dot{S}_L(s)}{S_L(s)} - \frac{\dot{X}_L^k(s)}{X_L^k(s)} \right) ds < x_k,$$

where $X_L^L \equiv Z_L$ and $X_L^G \equiv N_L$.

Proof. See Appendix B.4 for the formal proof. $\square$

Proposition 2 establishes the conditions under which as $t \rightarrow \infty$, $e_{i,H}^g(t) \rightarrow 1$ and $e_{i,L}^g(t), e_{i,G}^g(t) \rightarrow 0$. This proposition is used to close the model; the implied limiting case of the proposition is that the H -sector is the only active sector in the economy. Accordingly, this economy asymptotically hires only skilled workers and economy-wide TFP now grows at the constant growth rate g_H . Let $\tilde{x}(t)$ denote variable x in a steady state period $t \geq T$. The representative firm's profit-maximizing equilibrium condition is

$$\tilde{w}_H(t) = \tilde{p}_H(t)\tilde{A}_H(t). \quad (22)$$

With $\tilde{p}_H(t)$ as the numéraire, this pins down the equilibrium wage for skilled workers

$$\tilde{w}_H(t) = \tilde{A}_H(t). \quad (23)$$

To make the model stationary in steady state, I proceed in two steps. First, I change the timing index t to s to denote the stage of childhood development. This step acknowledges that while the generation problem is stationary, the behavior of parents — and therefore aggregate and equilibrium objects — may vary over the stages of childhood development. Second, I normalize steady state consumption and output by $\tilde{A}_H(s)$ to obtain $\tilde{c}_{i,H}^{g*}(s)$ and $\tilde{Y}_H^*(s)$, respectively. The stationarized version of the labor market clearing condition can be written as

$$\tilde{S}_H^d(s) = \mathcal{L} \int_h \int_{h'} h(1 - \tilde{\tau}(h, h'; s)) \tilde{\mu}(h, h'; s) dh' dh, \quad (24)$$

where the LHS is the demand for skilled labor and the RHS is the time-adjusted aggregate supply of skills in the economy in steady state stage s of childhood development. The corresponding sectoral good market clearing condition can be written as

$$\tilde{S}_H^s(s) = \mathcal{L} \int_h \int_{h'} \tilde{c}_H^*(h, h'; s) \tilde{\mu}(h, h'; s) dh' dh, \quad (25)$$

where the LHS is the normalized supply of H -sector output (i.e. $\frac{\tilde{Y}_H(s)}{\tilde{A}_H(s)}$) and the RHS is the normalized aggregate demand for H -sector output.

4 Mapping the Model to Data

This section discusses how the model is taken to the data. It starts by briefly discussing some assumptions underlying the numerical solution, and the remainder of this section lays out the calibration strategy and model fit. The parameters are disciplined using a combination of external calibration, direct measurement and internal calibration.

4.1 Numerical Solution

A detailed explanation of the solution algorithm is relegated to Appendix C.1. Instead, I elaborate here on two key assumptions that are required to proceed with the numerical solution. The first concerns the initial parent human capital distribution. This characterizes the model's initial conditions and is required to shoot forwards for the joint distribution. I

assume that, like the initial child human capital distribution, the initial parent human capital distribution is governed by a lognormal distribution $G(\eta, \kappa)$ where η and κ are the location and shape parameters, respectively.

The second key assumption is around timing. I assume that there are three stages of childhood development (i.e. $S = 3$) and that each stage of childhood development lasts for a single time period in the model such that ($S = T = 3$). As per the model's demographic structure, this means that each individual from a given generation lives $2T = 6$ periods in total, 3 as a child and 3 as a parent. This assumption is invoked without loss of generality and purely for computational tractability. This in turn implies that the human capital law of motion is governed by a total of four parameters including 1 human capital persistence parameter q and 3 parental time allocation elasticity parameters $\{\zeta_1, \zeta_2, \zeta_3\}$ where the subscripts denote the stage of childhood development. Empirically, I handle this by assuming that a single time period is equivalent to 5 years in the data.

4.2 External Calibration

The following 12 parameters are externally calibrated $\{\beta, \beta_g, \sigma, \chi, \epsilon, \phi_L, A_G(0), A_H(0), A_L(0), g_G, g_H, g_L\}$. Table 1 presents the values and sources of the externally calibrated parameters. The period discount factor β is assigned a value of 0.904. As in [Barro and Sala-i Martin \(2004\)](#), this assumes an annual discount factor of 0.98. Since a period in the model is 5 years, the discount factor is retrieved by 0.98^5 . I use an inter-generational discount factor β_g of 0.5 which is assumed by [Boar and Lashkari \(2021\)](#). Together, these values suggest that parents hardly attach value to the continuation value of their grandchildren. The elasticity of substitution of 0.1 suggests strong complementarities in consumption between the different sectoral goods which is a common finding in the structural change literature ([Herrendorf et al., 2013](#); [Buera and Kaboski, 2009](#)). I use the Frisch elasticity value of 0.5 from [Boppart et al. \(2024\)](#) to inform the value of χ .

The elasticity of substitution between skilled and unskilled workers in the L -sector production function ϵ is assigned a value of 2.9 implying gross substitutability between both types of labor. This value is obtained by [Acemoglu and Autor \(2011\)](#) who fit the method by [Katz and Murphy \(1992\)](#) to longer US time series data. The unskilled worker labor intensity of the L -sector denoted by ϕ_L is assigned a value of 0.75 to suggest that this sector uses unskilled labor more intensively than skilled labor. This closely matches the unskilled labor intensity in the US Low-Skill Services sector found in [Fang and Herrendorf \(2021\)](#).

Symbol	Description	Value	Source/Notes
<i>Panel A. Preferences</i>			
β	period discount factor	0.904	Barro and Sala-i Martin (2004)
β_g	inter-generational discount factor	0.500	Boar and Lashkari (2021)
σ	elast. of sub. b/n sectoral goods	0.100	Buera and Kaboski (2009)
χ	Frisch elasticity parameter	0.500	Boppart et al. (2024)
<i>Panel B. Technology</i>			
ϵ	L -sector elast. of sub. b/n skilled and unskilled	2.900	Acemoglu and Autor (2011)
ϕ_L	L -sector unskilled labor intensity	0.750	Fang and Herrendorf (2021)
$A_G(0)$	G -sector initial TFP level	1.000	normalization
$A_H(0)$	H -sector initial TFP level	1.000	normalization
$A_L(0)$	L -sector initial TFP level	1.000	normalization
g_G	G -sector average growth rate per period	0.053	Elbadri and Vizcaino (2025)
g_H	H -sector average growth rate per period	0.005	Elbadri and Vizcaino (2025)
g_L	L -sector average growth rate per period	0.037	Elbadri and Vizcaino (2025)

Table 1: Externally Calibrated Parameters

Notes: This table reports parameters that are calibrated using external evidence. The first column lists the mathematical symbol. The second column describes the parameter. The third column assigns the value for the parameter. The fourth column presents the source.

Sectoral TFP in the model grows exogenously according to an exponential growth function. All sectoral TFP levels are normalized to 1.0 in the initial period. The growth rates of sectoral TFP are obtained from the 44-industry version of the Elbadri and Vizcaino (2025) dataset which runs from 1947 to 2023. Broad sector TFP growth rates are calculated from the industry-level growth rates and aggregated using Törnqvist weights. The industry classification into broad sectors is consistent with Appendix A.1. The growth rates in Table 1 are per period (i.e. per 5 calendar years) and, unsurprisingly, show that the H -sector TFP grows at the slowest rate, followed by the L -sector.

4.3 Internal Calibration

There are 16 parameters left to calibrate. After normalizing θ_L to $1 - \theta_G - \theta_H$ and λ_L to 1, a total of 14 free parameters remain. These make up the 14-dimensional parameter vector

$$\Theta \equiv \{\theta_G, \theta_H, \lambda_G, \lambda_H, q, \zeta_1, \zeta_2, \zeta_3, \omega, \nu, \eta, \kappa, \eta_c, \kappa_c\}$$

The vector Θ is jointly identified to match patterns of structural change, time use and the skill distribution in postwar United States. The candidate vector that is chosen is the one that yields the minimum discrepancy between model-simulated and empirical data moments. This is accomplished via simulated method of moments (SMM).

At any given time, the model has just one cohort of parents and one cohort of children.

Reconciling cross-sectional data with model therefore requires building and tracking synthetic cohorts over time. I operationalize this by partitioning the sample period (1942-present) into six synthetic cohorts. This means that each calendar year is mapped to a unique combination of parent generation and childhood stage of development, as per the timing assumptions of the model. Each parent generation-child stage period lasts for a five-year window. More details on the timing convention can be found in Appendix C.2.

The list of targeted moments includes: (i) the share of skilled workers in the first, third and sixth generations, (ii) the log 90-10 wage per hour ratio of skilled workers in the first and sixth generations, (iii) the first and sixth generation High-Skill Services and Goods expenditure shares, (iv) the third and sixth generation parental childcare time allocation for parents with children in the first stage of childhood development (ages 0–4), (v) the fourth and sixth generation parental childcare time allocation for parents with children in the second stage of childhood development (5-9 years of age) and (vi) the third generation parental childcare time allocation for parents with children in the third stage of childhood development (10-14 years of age). Appendix C.3 specifies the data sources and details used to construct the 14 targeted data moments.

The choice of data moments is made such that the model reproduces patterns of structural change, time allocation and the skill distribution of the postwar US economy while simultaneously ensuring that the Jacobian matrix is of full rank. Heuristically, one can see this by splitting the parameters and moments into three blocks. The first block is concerned with matching the evolution of structural change in the US. The parameters that belong in this block are $\{\theta_G, \theta_H, \lambda_G, \lambda_H\}$. The consumption technology weights $\{\theta_G, \theta_H\}$ can be mapped to the initial aggregate expenditure shares while the parameters $\{\lambda_G, \lambda_H\}$ are particularly helpful in shaping final expenditure shares given initial shares by virtue of the fact that these two parameters control the income elasticity of demand.

The second block is concerned with matching time use patterns. The parameters that belong in this block are $\{\zeta_1, \zeta_2, \zeta_3, \omega, \nu\}$. The elasticities $\{\zeta_1, \zeta_2, \zeta_3\}$ control the importance of parental childcare time allocation in the human capital law of motion in each stage of childhood development which in turn shapes the continuation value that parents face. Cross-stage variation in childcare time allocation obtained from stage-parental childcare time allocation profiles can pin down stage-specific elasticities. Simultaneously, $\{\omega, \nu\}$ shape the weight and curvature of parental time allocation to childcare in flow utility, thereby determining the direct enjoyment that parents derive from time spent with children. These can be pinned down by the level and trend of childcare time allocation over generations.

The final block targets the supply of skills in the US. The parameters $\{\eta, \kappa, \eta_c, \kappa_c, q\}$ are placed in this block. The initial parent human capital distribution follows a lognormal

distribution $G(\eta, \kappa)$ where η and κ are the location and shape parameters, respectively. Conditional on the occupational choice threshold (which is endogenous), η and κ control the share of skilled workers and the variance of earnings conditional on being skilled (proxied by the log 90-10 wage gap conditional on being a skilled worker) in the first generation, respectively. Likewise, fixing the human capital production function parameters and the occupational choice threshold, the lognormal initial child human capital draw distribution $G(\eta_c, \kappa_c)$ can be disciplined by targeting the same moments of the sixth generation (current) US skill distribution. Finally, fixing the parental time investment elasticities in the human capital production function, the human capital persistence parameter q influences the extent to which a given level of parental time investments builds a comparative advantage in skilled work. It can therefore be disciplined by the skilled workers' share in between the initial and current US skill distributions.

Table 2 presents the values of the internally calibrated parameters. A worker disutility constant of 1.374 suggests that work is relatively distasteful for parents. The CRRA parameter value of 1.11 implies that parental time allocation is indeed treated as a luxury, with greater gross income generating a more than proportional increase in parental childcare time allocation. This value is however slightly lower than those obtained from similar papers that estimate this parameter in the context of time use such as [Bick et al. \(2022\)](#) and [Boppart et al. \(2024\)](#), but this is expected given that parental childcare time allocation is only one form of leisure. The SMM produces a higher G -sector consumption aggregator weight relative to the H - and L -sectors in order to match the higher expenditure share in this sector in initial periods. Looking at the income elasticity parameters and, coupled with the fact that sectoral goods are complements ($\sigma < 1$), it is clear that the H -sector is the income-elastic sector while the G -sector is the income-inelastic sector. Taken together, the preference parameter values suggest that as TFP rises, the two layers of income effects are simultaneously operating: households are biasing their consumption to H -sector while also increasing the parental childcare time allocation at the expense of work.

Symbol	Description	Value	Target/Notes
<i>Panel A. Preferences</i>			
ω	work disutility constant	1.374	childcare time allocation (stage 1, gen. 6)
ν	CRRA parameter	1.110	childcare time allocation (stage 2, gen. 6)
θ_G	G-sector consumption weight	0.575	G-sector expenditure share (gen. 1 av.)
θ_H	H-sector consumption weight	0.256	H-sector expenditure share (gen. 1 av.)
λ_G	G-sector income elasticity parameter	0.170	G-sector expenditure share (gen. 6 av.)
λ_H	H-sector income elasticity parameter	1.509	H-sector expenditure share (gen. 6 av.)
<i>Panel B. Human Capital Tech.</i>			
$1 - \zeta_1$	stage 1 parent time investment intensity	0.817	childcare time allocation (stage 1, gen. 3)
$1 - \zeta_2$	stage 2 parent time investment intensity	0.296	childcare time allocation (stage 2, gen. 4)
$1 - \zeta_3$	stage 3 parent time investment intensity	0.001	childcare time allocation (stage 3, gen. 3)
q	human capital persistence parameter	0.010	skilled workers share (gen. 3 av.)
<i>Panel C. Skill Distribution</i>			
η	initial parent human capital distr. (location)	0.321	skilled workers share (gen. 1 av.)
κ	initial parent human capital distr. (shape)	1.623	90-10 log wage gap of skilled workers (gen. 1 av.)
η_c	child human capital distr. (location)	0.399	skilled workers share (gen. 6 av.)
κ_c	child human capital distr. (shape)	1.000	90-10 log wage gap of skilled workers (gen. 6 av.)
<i>Panel D. Normalization</i>			
θ_L	L-sector consumption weight	0.169	backed out ($1 - \theta_G - \theta_L$)
λ_L	L-sector income elasticity parameter	1.000	fixed as reference for λ_G and λ_H

Table 2: Internally Calibrated Parameters

Notes: This table reports parameters that are calibrated internally. The 14 independent parameters are jointly identified via simulated method of moments (SMM). The remaining two parameters θ_L and λ_L are obtained via normalizations. The first column lists the mathematical symbol. The second column describes the parameter. The third column reports the parameter's value under the SMM best fit. The final column provides the most informative data moment used for identification.

Proceeding to the parameters that shape the human capital law of motion, it is clear that across stages of childhood development the importance of parental time investment decreases. In stage 1, the parental time intensity is 0.817 implying that this is the key driver of human capital accumulation for children when they are young. This number quickly falls in the second and third stages of childhood development to 0.296 and 0.001. This suggests a strong incentive for parents to front-load time investments in their children when they are young, especially since later stages become increasingly reliant on the human capital that is achieved then. This is in part identified by cross-stage variation in parental time investments. The falling parental time intensity resonates with much of the literature that stresses the importance of parental time allocation for early childhood human capital (Del Boca et al., 2014; Francesconi and Heckman, 2016; Yum, 2023; Bono et al., 2016). The fact that the human capital persistence reaches a lower-bound corner implies that all the variance in skilled worker wages observed in the data is inherited from the initial child distribution rather than persistence in the human capital production function. The skill distribution parameters provide the location and shape parameters

that are required to match the skill distribution in the initial and end generations.

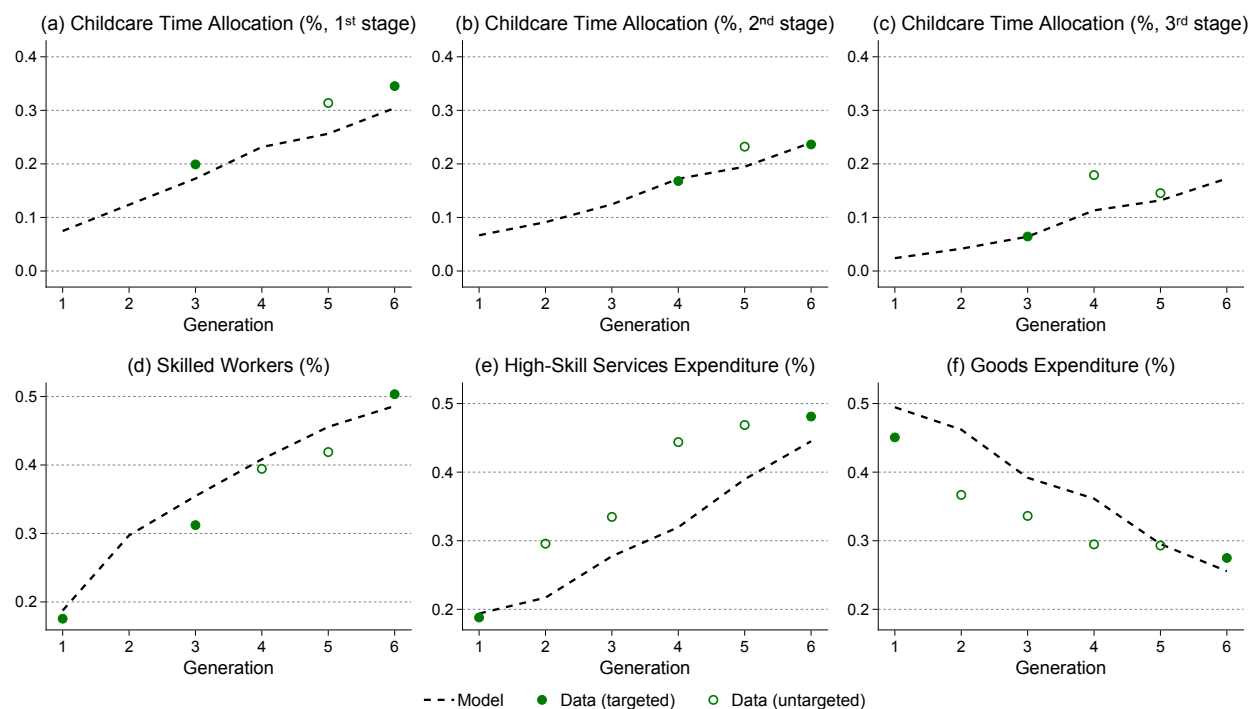


Figure 6: Model Fit — Data vs. Model-Simulated Moments

Notes: This figure plots the model-simulated moments obtained by the SMM with the data moments obtained for the US. Panel (a) plots the time share to childcare by parents with children in the first stage of childhood development (ages 0–4). Panel (b) plots the time share to childcare by parents with children in the second stage of childhood development (5–9 years old). Panel (c) plots the time share to childcare by parents with children in the third stage of childhood development (10–14 years old). Panel (d) plots the share of total employment in skilled occupations. Panel (e) plots the High-Skill Services expenditure share. Panel (f) plots the Goods expenditure share. Dashed lines are model-simulated moments while circles are data moments. Solid circles are data moments that are targeted by the SMM while hollow circles are data moments that are not targeted by the SMM.

Figure 6 shows the extent to which the model with the given parameter selection can replicate patterns of structural change, time allocation and the skill distribution in postwar United States. In all six panels, the x-axis is the generation. As per the timing convention in Appendix C.2, Generation 1 begins from 1942 and lasts until 1956 while Generation 6 lasts from 2017 to 2031. Solid red circles are targeted data moments, hollow red circles are untargeted data moments and the green lines are the model-simulated moments. Starting with parental time allocation, the model is able to replicate the cross-stage variation in both levels and changes in parental childcare time allocation (as a share of childcare and work time) over time as shown in the top row of Figure 6. This is particularly helpful for disciplining the stage-specific parental time elasticities in the human capital production function. Looking at the skill distribution, the model reproduces the skilled worker

shares in the initial and final generations. The generation 3 skilled workers share is slightly overstated in the model, explaining the lower-bound corner in the human capital persistence parameter q obtained by the SMM. Finally, the model is also able to replicate structural change in household expenditure away from the G -sector and into the H -sector over time.

5 Quantitative Exercises

This section performs four quantitative exercises using the calibrated version of the model. The first exercise quantifies the role of initial conditions — TFP in particular — in shaping structural change. The second and third exercises focus on quantifying the effect of early childhood development on structural change. Specifically, the second exercise assesses the role of the rise in parental time allocation observed in the US while the third exercise assesses the role of dynamic complementarity in shaping human capital outcomes and, by extension, structural change. The final exercise quantifies the role of human capital inequality in driving structural change.

5.1 Exercise 1 — Initial TFP Level and Structural Change

Motivation. The model’s initial conditions are defined by the initial parent human capital distribution and initial TFP levels (which are equalized across sectors). The latter is effectively the lever that controls the initial level of development of the economy and, together with the sectoral TFP growth rates, is the source of income effects that shape both supply- and demand-side drivers of structural change. To highlight these mechanics, this counterfactual exercise explores how initial TFP influences structural change.

Implementation. Besides the baseline simulation, two counterfactual simulations are run. The first equalizes initial TFP levels across sectors j at 0.75 (i.e. $A_j(0) = 0.75$) while the second equalizes them at 1.25 (i.e. $A_j(0) = 1.25$) compared to their baseline values of $A_j(0) = 1.00 \forall j \in \{G, L, H\}$. For context, the 25 percentage point gap in TFP is broadly in line with the US-Zimbabwe and US-South Africa TFP gap in the 1960s according to the Penn World Tables (Feenstra et al., 2015). All simulations keep the initial parent human capital distribution and TFP growth rates fixed at their baseline values.

Results. The results of the first exercise are presented in Figure 7. The key takeaway is that initial TFP levels govern the speed of the transitional dynamics. The top-left panel plots the generation-average H -sector expenditure share and shows that for any given

generation, a higher initial TFP level raises the *H*-sector expenditure share. For generation 1, a 25 percentage point increase (decrease) in initial TFP level relative to baseline increases (decreases) the *H*-sector expenditure by 5.9 (7.4) percentage points relative to the baseline. These level differences are largely persistent; by generation 6, the counterfactual scenario where initial TFP level was 25 percentage points higher (lower) relative to baseline sees a higher (lower) *H*-sector expenditure by 4.7 (6.1) percentage points relative to baseline. Another representation of this outcome is recorded in the middle-left panel which plots the number of generations required for the generation-average *H*-sector expenditure share to exceed 40% against the initial TFP level. The higher the initial TFP level, the fewer the number of generations that are required for the *H*-sector expenditure share to exceed the 40% threshold. These results square with the positive link between the High-Skill Services employment share and the level of development shown in Figure 2.

The top-right panel of Figure 7 shows that together with the rise in the High-Skill Services expenditure share, higher initial TFP levels induce a rise in the share of skilled workers in the economy. Two mechanisms are at play. The first is the rise in the demand for skills. Fixing the parent human capital distribution, higher TFP levels across sectors enrich households. By doing so, households bias their consumption towards the *H*-sector given that this is the income-elastic sector. With the *H*-sector being the skill-intensive sector, this raises the demand for skills in the economy. Reinforcing this mechanism is the effect of differential TFP growth rates; slow TFP growth in the *H*-sector increases the wage of skilled workers in the *H*-sector, thereby raising the returns to skills. With perfect foresight, parents internalize these dynamics and invest more time in their children given their altruistic nature. The second mechanism highlights the role of income effects in driving greater time investments in their own right. Non-homothetic preferences in the consumption-time allocation margin implies that as households get richer, they invest more time in their children because they enjoy doing so. With greater investments, next-generation parents enjoy greater human capital, effectively making them richer not only because of counterfactually higher TFP levels, but also because of higher human capital levels. Thus, parents effectively pass on these two dynamics to further generations, consequently driving persistent differences in structural change as a result of differences in initial conditions.

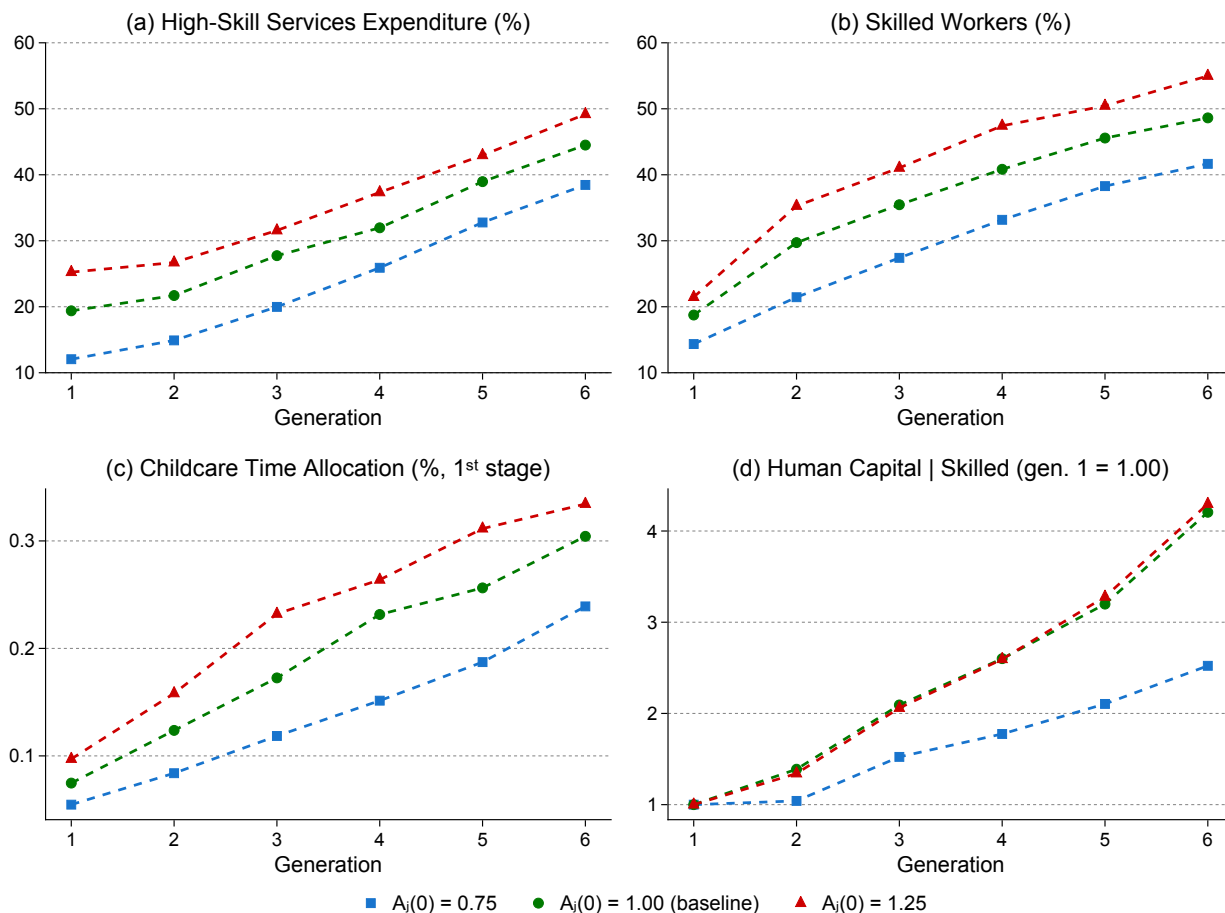


Figure 7: Exercise 1 — The Role of Initial TFP Levels

Notes: This figure plots the results from the first counterfactual exercise which aims to quantify the effect of initial TFP levels on structural change. Panel (a) plots the generation-average expenditure share of High-Skill Services against generations. Panel (b) plots the generation-average share of skilled workers against generations. Panel (c) plots the childcare time allocation of parents towards young children (ages 0–4). Panel (d) plots the average human capital level conditional on being a skilled worker. Circles denote the baseline simulation with initial sectoral TFP normalized to 1.00 for all sectors. Triangles and squares denote the simulations with initial sectoral TFP set to 1.25 and 0.75, respectively.

These dynamics can be verified by the rise in childcare time allocation by parents towards their children when their children are youngest. The bottom-left panel of Figure 7 shows that higher initial TFP levels drive greater time investments not just by the initial generation, but also subsequent generations. The 50 percentage point gap in initial TFP levels generates just a 4.3 percentage points in time allocation shares in the first generation but a 9.5 percentage point difference in the sixth generation. The bottom-right panel shows that these investments materialize into a far more human capital-intensive economy.

5.2 Exercise 2 — Childcare Time Allocation and Structural Change

Motivation. What role does early childhood development play in driving structural change? Motivated by the importance of parental time allocation as an input in the human capital development of young children (Del Boca et al., 2014; Yum, 2023) and the finding in Section 4 that the human capital of children in the first stage of childhood development (i.e. between 0 and 4 years old) is particularly sensitive to parental time inputs, this section quantifies the historical rise in the childcare time allocation of parents towards young children on structural change in the US.

Implementation. This exercise is implemented by fixing the childcare time allocation of all parents with children in the first stage of childhood development (i.e. between 0 and 4 years old) to the generation 1 weighted average value of 7.5% for all subsequent generations. Parents can therefore still make time allocation choices, but only when their children are in the second and third stages of childhood development. These are now conditional on the fixed time input that parents allocate to childcare when the children are youngest. To isolate the effects of the skill distribution, this exercise distinguishes between partial and general equilibrium. The former freezes relative prices to their baseline values while the latter allows relative prices to evolve such that market clearing is achieved.

Results. Table 3 reports the results of this counterfactual exercise. The first column reports the moment of interest. The next two columns report the values from the baseline simulation for the first and sixth generations. The subsequent two columns report the partial equilibrium simulation results (i.e. fixing relative prices and wages to baseline values). The final two columns report the general equilibrium simulation results (i.e. allowing relative prices to evolve such markets clear). The headline result is that the rise in parental time allocation towards young children (i.e. 0-4 years old) can explain 64.3% of US structural change to *H*-sector in partial equilibrium and 35.0% in general equilibrium.

Starting with the partial equilibrium result, the aggregate *H*-sector expenditure share rises from 21.5% to just 30.5% as opposed to from 19.4% to 44.5% in the baseline case between generations 1 and 6. Panel B reveals that this is driven by the sharp decline in skills. With relative prices fixed, the skilled worker threshold across generations is unchanged. Lower parental time investments in young children mechanically compresses the supply of human capital across generations. With the human capital distribution shifting to the left, fewer workers have a comparative advantage in skilled work. Consequently, the share of skilled workers falls to 11.9% by generation 6 compared to 48.6% in the baseline simulation implying a considerable decline in the supply of skilled workers in the economy. Clearly however, the decline in the supply of skills does not only happen through a fall

in the quantity of skilled workers but also through the quality of skilled workers; among skilled workers, the average human capital level among sixth-generation parents falls by almost 50%. Both the quality and quantity margins therefore have a meaningful role in restricting output in H -sector by virtue of the fact that this is the skill-intensive sector.

	Baseline		PE		GE	
	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
<i>Panel A. Time Allocation</i>						
Childcare time allocation share (0–4 yrs old)	0.075	0.304	0.075	0.075	0.075	0.075
Childcare time allocation share (5–9 yrs old)	0.067	0.239	0.058	0.146	0.059	0.206
Childcare time allocation share (10–15 yrs old)	0.024	0.173	0.024	0.137	0.024	0.148
<i>Panel B. Human Capital (HK)</i>						
Skilled work threshold (gen. 1 = 1.00)	1.000	3.649	1.000	3.649	1.000	1.555
Skilled workers share	0.187	0.486	0.187	0.119	0.177	0.468
Mean HK HK > skilled threshold (gen. 1 = 1.00)	1.000	4.206	1.000	2.148	1.000	0.992
<i>Panel C. Structural Change</i>						
H -sector expenditure share	0.194	0.445	0.215	0.305	0.205	0.368
<i>Panel D. Intergenerational Change (gen. 6 – gen. 1)</i>						
ΔH -sector expenditure share	0.251		0.090		0.163	
Share of baseline	1.000		0.357		0.650	
<i>Panel E. Welfare</i>						
Consumption equivalent variation (%)	0.000	0.000	–5.033	–39.142	–6.425	–56.960

Table 3: Exercise 2 — Quantifying the Effects of Parental Time Investments in Childcare

Notes: This table reports the results of the second counterfactual exercise which quantifies the contribution of increased parental time investment in childcare during the first stage of childhood (ages 0–4) to structural change. In the counterfactual, parental childcare time during this stage is held fixed at its generation 1 baseline average of 7.5% in all generations. Panel A reports parental childcare time allocation. Panel B reports moments describing the distribution of human capital (HK). Panel C reports the H -sector expenditure share. Panel D reports the change in the H -sector expenditure share between generations 1 and 6 and expresses that change as a share of the corresponding baseline change. Panel E reports the ex-ante dynastic consumption equivalent variation relative to the baseline. The first two numerical columns report generation 1 and generation 6 outcomes in the baseline simulation. The next two columns report outcomes under the partial-equilibrium (PE) counterfactual, in which relative prices and wages are held fixed at their baseline values. The final two columns report outcomes under the general-equilibrium (GE) counterfactual, in which relative prices and wages adjust to clear markets.

Indirectly, the sharp decline in the supply of skills in the economy means that demand for H -sector output is also compressed. With parents effectively poorer because of the decline in human capital levels, non-homothetic preferences are more muted compared to the baseline simulation. This force outweighs any potential increases in income coming from endogenously allocating more time at work. Demand for output from the H -sector (which is the income elastic sector) therefore declines. Relative price effects are unchanged

in the partial equilibrium simulation, so the demand-side effects operate only through non-homothetic preferences.

The aggregate welfare implications are reported in Panel E; the distribution-weighted consumption equivalent variation in partial equilibrium implies that in the first generation, parents and their descendants must take a 5.0% reduction in per-period baseline consumption in order to be indifferent between the baseline and the partial equilibrium counterfactual. The decline is sharper for parents and their descendants in the sixth generation (39.1%) now that their human capital has reacted endogenously to the freeze in parental time allocation of young children.

In the general equilibrium simulation, the aggregate *H*-sector expenditure share rises from 20.5% in generation 1 to 36.8% in generation 6 as opposed to from 19.4% to 44.5% in the baseline scenario, respectively. Accordingly, when relative prices and wages are allowed to clear markets in the counterfactual simulation, the rise in US parental time allocation towards the childcare of young children can explain 35.0% of structural change to the *H*-sector. Evolving relative prices lead to a sharp decline in the threshold of skilled work from 3.65 in the baseline simulation to 1.56 in the counterfactual general equilibrium simulation by generation 6, so while the skill distribution still shifts to the left, more workers can access skilled work relative to the partial equilibrium case because of the lower threshold. Consequently, the skilled workers share falls by only 1.8 percentage points in generation 6 relative to the baseline.

Now, however, with parents that have lower human capital accessing skilled work, this weighs down the average human capital level of skilled workers to the extent that it is largely unchanged across six generations. Thus, despite the recovery of the skilled workers shares in the economy, the aggregate welfare implications are larger in magnitude than in the partial equilibrium counterfactual, with the consumption equivalent variation now -6.4% (-57.0%) for first- (sixth-) generation parents and their descendants. The core insight of this exercise is that US parental time investments towards young children has served primarily as a human capital quality amplifier, operating mainly through the quality margin rather than the quantity margin.

5.3 Exercise 3 — Dynamic Complementarity and Structural Change

Motivation. Incorporating the multi-stage human capital production function (6) into a multi-sector growth model is one of the innovations of this paper. A central feature that the production function captures is the notion of dynamic complementarity; the productivity of an additional unit of investment is increasing in the level of human capital that a child has already acquired at any given time. The cumulative nature of the human capital accumulation process makes a strong case for the importance of early childhood

development in explaining the supply of skills in the economy. The aim of this quantitative exercise is to assess numerically how important these incentives are in shaping investment decisions and how these decisions themselves explain the sectoral make-up of a given country.

Implementation. This exercise is implemented by modifying the human capital production function (6). Now, the human capital production will only feature the human capital draw of a child at birth in all stages of childhood development rather than the current human capital of the child, implying the following setup

$$h_i^{g+1}(t+1) - h_i^{g+1}(t) = h_i^{g+1}(gT)^{\zeta_s+q} \left(h_i^g \tau_i^g(t) \right)^{1-\zeta_s}, \quad (26)$$

such that the cross-partial derivative $\frac{\partial^2 h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t) \partial \tau_i^g(t)} = 0$. Intuitively, (26) suggests that the human capital that a child has acquired in their early childhood no longer shapes parental time investment decisions in subsequent stages of childhood development. As a result, any source of dynamic complementarity here comes from the human capital draw at birth rather than the human capital that the child has accumulated. As with the second counterfactual exercise, this exercise distinguishes between partial and general equilibrium. The former freezes relative prices at baseline values while the latter allows relative prices and wages to clear markets.

Results. The results of this counterfactual exercise are presented in Table 4. The first column reports the moment of interest. The next two columns report the values from baseline simulation for the first and sixth generations. The subsequent two columns report the partial equilibrium simulation results (i.e. fixing relative prices and wages to baseline values). The final two columns report the general equilibrium simulation results (i.e. allowing relative prices to evolve such markets clear). The headline result is that dynamic complementarity stemming from past human capital investments from parents can explain 64.0% of US structural change to H -sector in partial equilibrium and 33.1% in general equilibrium with aggregate welfare implications similar in magnitude to the second counterfactual exercise.

Aside from the headline result, the mechanics also resemble those in the second exercise. In the partial equilibrium case, the aggregate H -sector expenditure share rises from 20.4% to 29.5% compared to the baseline values of 19.4% and 44.5% across the six generations. Again, this is mostly driven by a decline in skills, with both the quality and quantity margins playing a role. The share of skilled workers in the economy falls from 48.6% in the baseline simulation to 4.7% in the counterfactual simulation, while the

average skill level among skilled workers declines by almost a half in the same generation. Now, however, parental time allocated to childcare declines across all stages of childhood development, both in levels and in changes across generations. This pattern suggests that shutting down dynamic complementarity across stages substantially weakens parents' incentives to invest in their children. The decline in the skills supply in the economy naturally reinforces this mechanism since the marginal benefit of time investments depend on the human capital of parents. Simultaneously, income effects that would shape the time allocation problem that parents face are now more muted relative to the baseline benchmark.

	Baseline		PE		GE	
	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
<i>Panel A. Time Allocation</i>						
Childcare time allocation share (0–4 yrs old)	0.075	0.304	0.039	0.079	0.046	0.181
Childcare time allocation share (5–9 yrs old)	0.067	0.239	0.035	0.136	0.040	0.151
Childcare time allocation share (10–15 yrs old)	0.024	0.173	0.024	0.134	0.024	0.153
<i>Panel B. Human Capital (HK)</i>						
Skilled work threshold (gen. 1 = 1.00)	1.000	3.649	1.000	3.649	1.000	1.090
Skilled workers share	0.187	0.486	0.187	0.047	0.185	0.415
Mean HK HK > skilled threshold (gen. 1 = 1.00)	1.000	4.206	1.000	2.142	1.000	0.689
<i>Panel C. Structural Change</i>						
<i>H</i> -sector expenditure share	0.194	0.445	0.204	0.295	0.200	0.368
<i>Panel D. Intergenerational Change (gen. 6 – gen. 1)</i>						
Δ <i>H</i> -sector expenditure share	0.251		0.091		0.168	
Share of baseline	1.000		0.360		0.669	
<i>Panel E. Welfare</i>						
Consumption equivalent variation (%)	0.000	0.000	–5.894	–41.421	–11.709	–66.914

Table 4: Exercise 3 — Quantifying the Effects of Dynamic Complementarity

Notes: This table reports the results of the third counterfactual exercise which quantifies the contribution of dynamic complementarity in the human capital production function to structural change. The exercise replaces the baseline multi-stage human capital production function in equation (6) with the counterfactual specification in equation (26). Panel A reports parental childcare time allocation. Panel B reports moments describing the distribution of human capital (HK). Panel C reports the *H*-sector expenditure share. Panel D reports the change in the *H*-sector expenditure share between generations 1 and 6 and expresses that change as a share of the corresponding baseline change. Panel E reports the ex-ante dynastic consumption equivalent variation relative to the baseline. The first two numerical columns report generation 1 and generation 6 outcomes in the baseline simulation. The next two columns report outcomes under the partial-equilibrium (PE) counterfactual, in which relative prices and wages are held fixed at their baseline values. The final two columns report outcomes under the general-equilibrium (GE) counterfactual, in which relative prices and wages adjust to clear markets.

In the general equilibrium case, the *H*-sector expenditure share rises from 20.0% to 36.8%. Now, relative prices and wages evolve to clear markets, reshaping the returns

to skills and, consequently, incentives to invest in human capital. The skilled workers share rises to reach 41.5% by the sixth generation as a result of the decline in the skilled workers threshold which depends on relative prices, effectively weakening the role of the quantity margin. Like the second counterfactual, the average human capital of skilled workers declines by more than in the partial equilibrium case since less capable workers can now access skilled work. Fundamentally, this counterfactual exercise underscores the importance of dynamic complementarity across stages of childhood development in shaping the aggregate supply of skills in the economy. An important implication of this is that since parental time investment elasticity are largest — by a considerable margin — when children are youngest, early childhood development through dynamic complementarity has been an important engine that has shaped the sectoral composition of aggregate output in the US.

5.4 Exercise 4 — Human Capital Inequality and Structural Change

Motivation. The distribution of human capital is key for the dynamics of the theory. It maps directly into the demand for different sectoral goods as well as the supply of skills in the economy. Non-linear technologies — including non-homothetic preferences and dynamic complementarity in the human capital accumulation technology — imply that the relevant moments of the distribution go beyond just the simple mean and, as a result, the forces that shape structural change will depend in part on the extent of dispersion of human capital in the economy as well as the mean. The purpose of this quantitative exercise is to understand and quantify the role of human capital inequality in shaping structural change.

Implementation. The two parameters that control the extent of human capital inequality in the economy are the initial parent human capital and initial child human capital distribution shape parameters κ and κ_c , respectively. This quantitative exercise simultaneously scales both, effectively modifying the rate of dispersion

$$\hat{\kappa} = \psi\kappa \quad \text{and} \quad \hat{\kappa}_c = \psi\kappa_c \quad \forall \psi \in \{0.65, 1.00, 1.35\} \quad (27)$$

However, since increasing the rate of dispersion would increase the lognormal average level of human capital, I also change the location parameters η and η_c such that the means are preserved at their baseline levels. Mean preservation requires that the new counterfactual location parameter take on the values

$$\hat{\eta} = \eta + \frac{\kappa^2 - \hat{\kappa}^2}{2} \quad \text{and} \quad \hat{\eta}_c = \eta_c + \frac{\kappa_c^2 - \hat{\kappa}_c^2}{2} \quad (28)$$

This ensures that any changes in the simulation results are attributed solely to the change in the distribution of human capital rather than the average.

Results. The results are presented in Table 5. The first column reports the moment of interest. The next two columns report the generation 1 and generation 6 average values from the low inequality simulation where the parameters κ and κ_c are scaled by 0.65 relative to the baseline. The subsequent two columns report the generation 1 and generation 6 average values from the baseline simulation. The final two columns report the generation 1 and generation 6 average values from the high inequality simulation where the parameters κ and κ_c are scaled by 1.35. The key takeaway is that while raising the scaling factor from 0.65 to 1.35 raises the starting average H -sector expenditure share from 14.6% to 26.1%, it slows down the increase in the H -sector expenditure share from 27.7 to 20.3 percentage points between the first and sixth generation.

Starting with the supply side, Panel A reveals that time allocated to childcare — particularly towards younger children — is falling with inequality. This is driven by two forces. First, complementarities in the human capital production function with parent human capital implies that parents have to compensate for the decline in parent human capital by investing more time in their children. Parents in the thicker upper tail in the high inequality simulation can afford to spend less time with their children while simultaneously ensuring that they can access skilled work when they enter the labor market. Second, a more compressed initial child human capital distribution implies that skilled work is more accessible for the larger share of children in exchange for a relatively smaller opportunity cost for parents. Parents take advantage of this by investing more time in their children given that gross income is convex in human capital.

Panel B shows that the initial skilled workers share falls with inequality, which is a natural result of the fact that a more compressed parent human capital distribution is more likely to lead to a higher share of parents exceeding the threshold. By generation 6, the skilled workers share also falls with inequality. This coincides with the fact that more compressed distributions lead to a broad-based increase in parental time allocation towards childcare by parents with young children. This, together with the fact that the skilled work threshold grows fastest in the low inequality case, explains the dramatic rise in the average skill level among skilled workers in the low inequality case relative to the high inequality case and the consequent increase in the distribution-weighted consumption equivalent variation. Thus, high inequality dampens incentives for human capital quality gains, contributing to the slowdown in the rate of structural transformation.

	Low Ineq. ($\psi = 0.65$)		Baseline ($\psi = 1.00$)		High Ineq. ($\psi = 1.35$)	
	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
<i>Panel A. Time Allocation</i>						
Childcare time allocation share (0–4 yrs old)	0.089	0.312	0.075	0.304	0.057	0.285
Childcare time allocation share (5–9 yrs old)	0.073	0.239	0.067	0.239	0.057	0.233
Childcare time allocation share (10–15 yrs old)	0.022	0.171	0.024	0.173	0.024	0.171
<i>Panel B. Human Capital (HK)</i>						
Skilled work threshold (gen. 1 = 1.00)	1.000	3.837	1.000	3.649	1.000	3.425
Skilled workers share	0.224	0.494	0.187	0.486	0.145	0.473
Mean HK HK > skilled threshold (gen. 1 = 1.00)	1.000	5.593	1.000	4.206	1.000	1.924
<i>Panel C. Structural Change</i>						
H-sector expenditure share	0.146	0.424	0.194	0.445	0.262	0.465
<i>Panel D. Intergenerational Change (gen. 6 – gen. 1)</i>						
Δ H-sector expenditure share		0.277		0.251		0.203
Share of baseline		1.104		1.000		0.807
<i>Panel E. Welfare</i>						
Consumption equivalent variation (%)	2.789	10.069	0.000	0.000	–9.559	–16.577

Table 5: Exercise 4 — Quantifying the Effects of Human Capital Inequality

Notes: This table reports the results of the fourth counterfactual exercise which quantifies the role of human capital inequality in shaping structural change. The exercise jointly varies the shape parameters κ and κ_c of the lognormally distributed initial parental and child human capital draws. The corresponding location parameters, η and η_c , are adjusted to preserve the means of the two distributions. The parameter ψ denotes the scaling factor applied to κ and κ_c : $\psi = 0.65$ represents the low-inequality counterfactual, $\psi = 1.00$ represents the baseline, and $\psi = 1.35$ represents the high-inequality counterfactual. Panel A reports parental childcare time allocation. Panel B reports moments describing the distribution of human capital (HK). Panel C reports the H -sector expenditure share. Panel D reports the change in the H -sector expenditure share between generations 1 and 6 and expresses that change as a share of the corresponding baseline change. Panel E reports the ex-ante dynastic consumption equivalent variation relative to the baseline. Within each specification, the two columns report average outcomes for generations 1 and 6, respectively.

On the demand side, the increasing initial H -sector expenditure share in inequality is a direct artifact of a thicker right tail in the initial parent human capital distribution and non-homothetic sectoral preferences; richer parents with higher human capital initially earn enough to front-load demand for H -sector output. The decline in the rate of structural change, on the other hand, results from concavity in the Engel curve. Higher mean-preserving inequality in human capital implies a larger share of parents sit on the higher-end of the human capital distribution, where gains in income translate into smaller rises in the H -sector expenditure share relative to parents on the lower-end of the distribution. Taken together, these forces explain the declining rate of structural change to the H -sector with human capital inequality.

6 Conclusion

This paper proposes a novel theory whereby early childhood development shapes structural change. I motivate my theory with empirical evidence that the High-Skill Services share, parental time allocation to younger children and early childhood human capital outcomes all rise with development. I then build a multi-sector heterogeneous agents model in general equilibrium to rationalize these facts. In the model, parents are altruistic and enjoy spending time with their children. As sectoral TFP rises, non-homothetic sectoral preferences generate an increased demand for the income-elastic sector (High-Skill Services). Simultaneously, income effects and anticipation of increasing return to skills induce parents to invest more time in their children in early stages of their childhood. Through dynamic complementarity, this increases the supply of skilled workers in the economy which benefits High-Skill Services disproportionately given that this sector employs skills more intensively. Endogenizing supply- and demand-side forces means that the model can account for the feedback effects between both forces.

After calibrating the model to match patterns of structural change, time use and the skills distribution over time in the US, I use the quantitative version of the model to perform several counterfactual exercises. The first quantifies the role of initial TFP levels — the main development lever — in shaping structural change. The second exercise measures the effect of the rise in parental time allocation to childcare of young children across generations on US structural change while the third exercise explores the role of dynamic complementarity in shaping time investments and, by extension, structural change. The final exercise measures the role of human capital inequality in driving structural change.

These results generate key policy implications. The first is that in the context of limited industrialization opportunities and the growing importance of Services in both advanced and developing countries, human capital will prove central for development. Second is that policymakers should not treat human capital as a one-step process; human capital accumulation is not only multi-stage, but also a process where human capital accumulation across stages of childhood development is subject to dynamic complements. This means that early childhood human capital investment amplifies incentives for human capital accumulation in later stages. Policies to improve early childhood human capital are likely to generate large gains for structural change and, ultimately, development going forward.

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A Additional Empirics

A.1 Industry Classification

Table A.1 presents the industry classification for Services. The remaining industries make up the Goods sector. This includes Agriculture, Fishing and Forestry, Mining and Extraction, Manufacturing, Construction and Electricity, Gas, Water and Waste Management.

Sector	Industry	1960	1970	1980	1990	2000	2005	2010	2015	2020	1960-2020
High-Skill Services	Education	0.69	0.64	0.65	0.66	0.69	0.77	0.76	0.77	0.78	0.71
	Health and Social Work	0.52	0.47	0.48	0.54	0.56	0.55	0.53	0.70	0.77	0.57
	Financial Services and Insurance	0.40	0.40	0.45	0.55	0.59	0.69	0.72	0.53	0.59	0.55
	Business Services and Real Estate	0.46	0.44	0.49	0.50	0.52	0.54	0.53	0.63	0.68	0.53
Low-Skill Services	Other Services	0.32	0.33	0.42	0.47	0.45	0.39	0.33	0.23	0.25	0.35
	Public Administration and Defense	0.15	0.19	0.26	0.29	0.35	0.40	0.37	0.39	0.42	0.31
	Wholesale and Retail Trade	0.20	0.17	0.22	0.26	0.27	0.29	0.29	0.13	0.15	0.22
	Transportation, Storage, and Communications	0.10	0.10	0.14	0.17	0.18	0.14	0.29	0.28	0.28	0.19
	Hotels and Restaurants	0.17	0.14	0.15	0.17	0.14	0.20	0.19	0.13	0.14	0.16
	Private Household Services	0.03	0.01	0.03	0.03	0.02	0.01	0.01	0.03	0.21	0.04

Table A.1: Industry Classification for Services

Notes: This table is created using US census microdata in IPUMS-I as the benchmark for industry classification. Each column specifies the person-weighted share of total employment in the US coming from skilled occupations in a given INDGEN industry in the corresponding year. Skilled occupations are defined as observations whose primary occupation as per the OCCISCO variable is recorded as either legislator/senior official/manager, professional or technician/associate professionals. The underlying data keeps observations within the age range of 16 and 64 (inclusive) with an activity status employed as per the EMPSTAT variable. The final column computes the sample average between 1960 and 2020.

A.2 Parental Time Investments in Childcare

Figure A.1 restricts attention to diarists who live in a household with a child below the age of 5 and that this is the only child in the household.

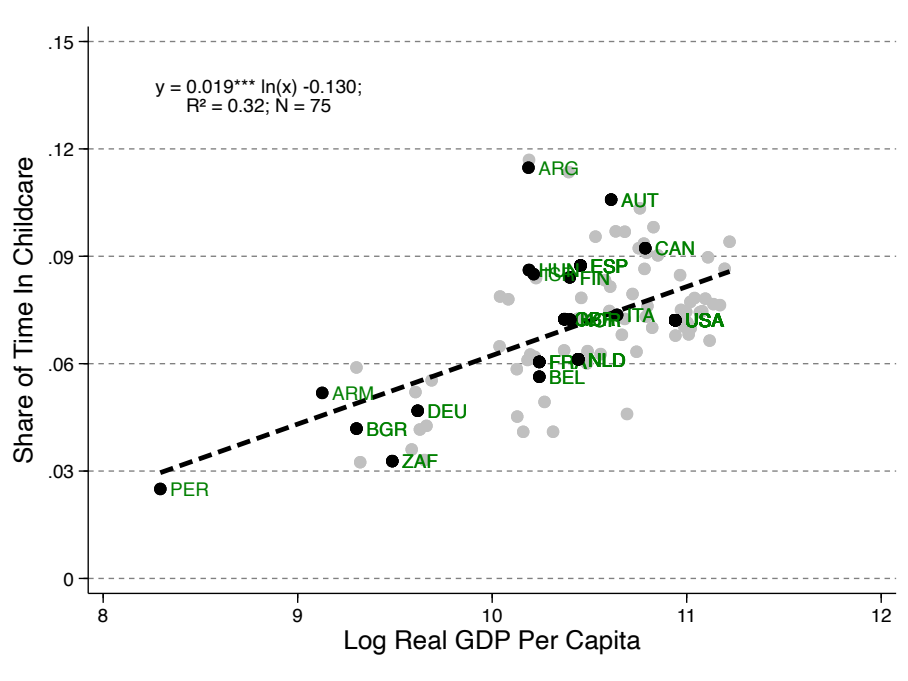


Figure A.1: Share of Time Spent on Childcare (Only Child Households)

Notes: The underlying data restricts attention to individuals aged between 18 and 60 (inclusive). Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights to correct for differences in days of the week. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the share of time spent on childcare on log PPP-adjusted real GDP per capita using the country-year observations. ** *** statistically significant at the 10% (5%) [1%] level.

A.3 Parental Time Investments in Childcare Using IPUMS-MICS

The IPUMS-MICS dataset surveys the following six binary questions for both mothers and fathers of children aged between 0 and 4 years old:

- Read books to the child in the last 3 days
- Told stories to child in the last 3 days
- Sang songs to the child in the last 3 days
- Took child outside in the last 3 days
- Played with the child in the last 3 days
- Spent time with child naming, counting, drawing things in the last 3 days

Aggregating the binary answers (1 for yes, 0 for no) for each household and aggregating at the country-year level using the relevant survey weights yields the parental time investment in child index. This is plotted against the log of real GDP per capita in Figure [A.2](#).

Figure A.2: Parental Time Investment in Childcare Index (Using IPUMS-MICS)

Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed under-5 child survey weights. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the parental time investment in child index on log PPP-adjusted real GDP per capita using the country-year observations. *(**)[***] statistically significant at the 10% (5%) [1%] level.

A.4 Parental Time Investments in Childcare of Younger Children in US

Figure A.3 isolates US time use data from 1965 to 2023 shows that, with the exception of 1975, there is a consistent increase in time spent on childcare over time.

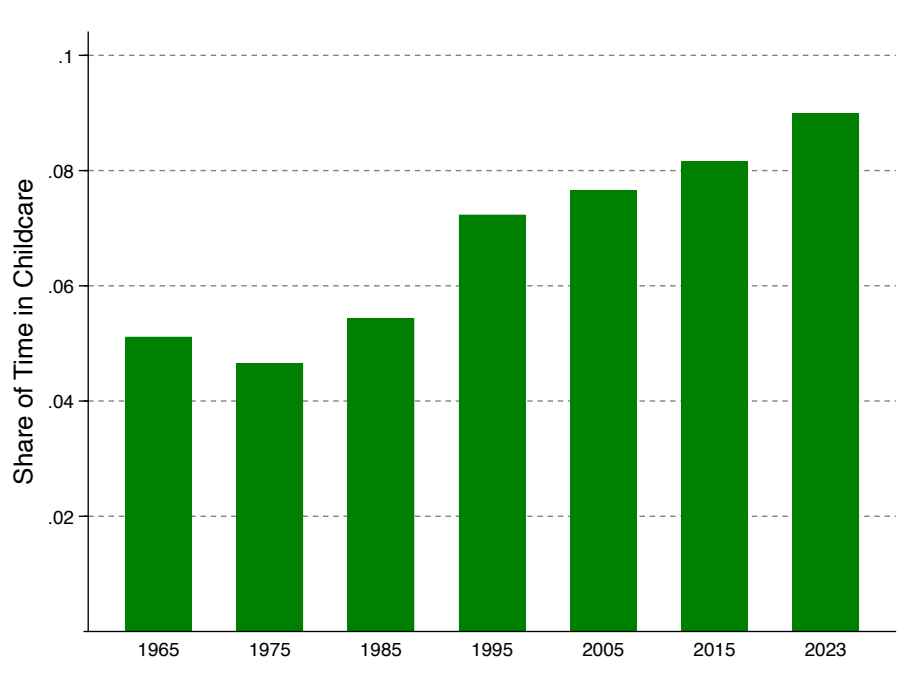


Figure A.3: Share of Time Spent on Childcare Over Time in the US

Notes: The underlying data restricts attention to individuals aged between 18 and 60 (inclusive).

A.5 Parental Time Investments in Childcare of Younger Children By Gender Using IPUMS-MICS

Using IPUMS-MICS data and following the same strategy in Appendix A.3, I plot the parental time investment index by the gender of the parent in Figure A.4. The link shows a very similar trend to Figure 4 — both men and women spend more time with their offsprings with a semi-elasticity that is statistically significant at the 1% level, and that there are qualitatively import level differences between men and women.

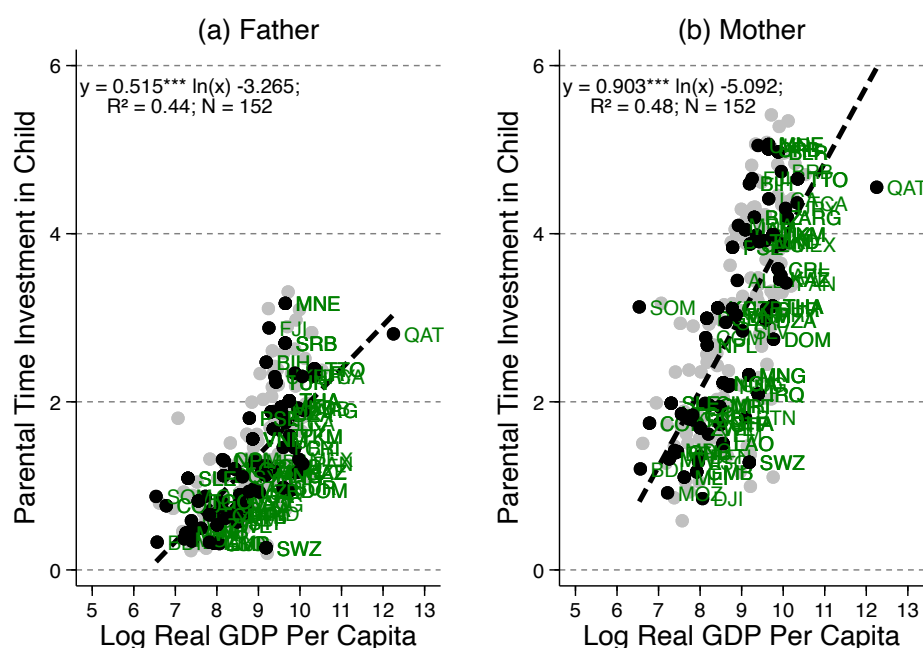


Figure A.4: Parental Time Investment in Childcare Index by Gender (Using IPUMS-MICS)

Notes: Panel (a) plots the parental time investment in childcare index for fathers. Panel (b) plots the parental time investment in childcare index for mothers. Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed under-5 child survey weights. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the parental time investment in child index on log PPP-adjusted real GDP per capita using the country-year observations. ^{*}(^{**})[^{***}] statistically significant at the 10% (5%) [1%] level.

A.6 Female Time Investments in Childcare of Younger Children By Employment Status

Using time use data, I plot the time allocated to the childcare of young children by females by employment status. Figure A.5 shows that the time allocation trend in Figure 4 is robust to female labor force participation trends, given that the correlation is positive and statistically significant at the 1% level for full-time and part-time employed females as well as unemployed females.

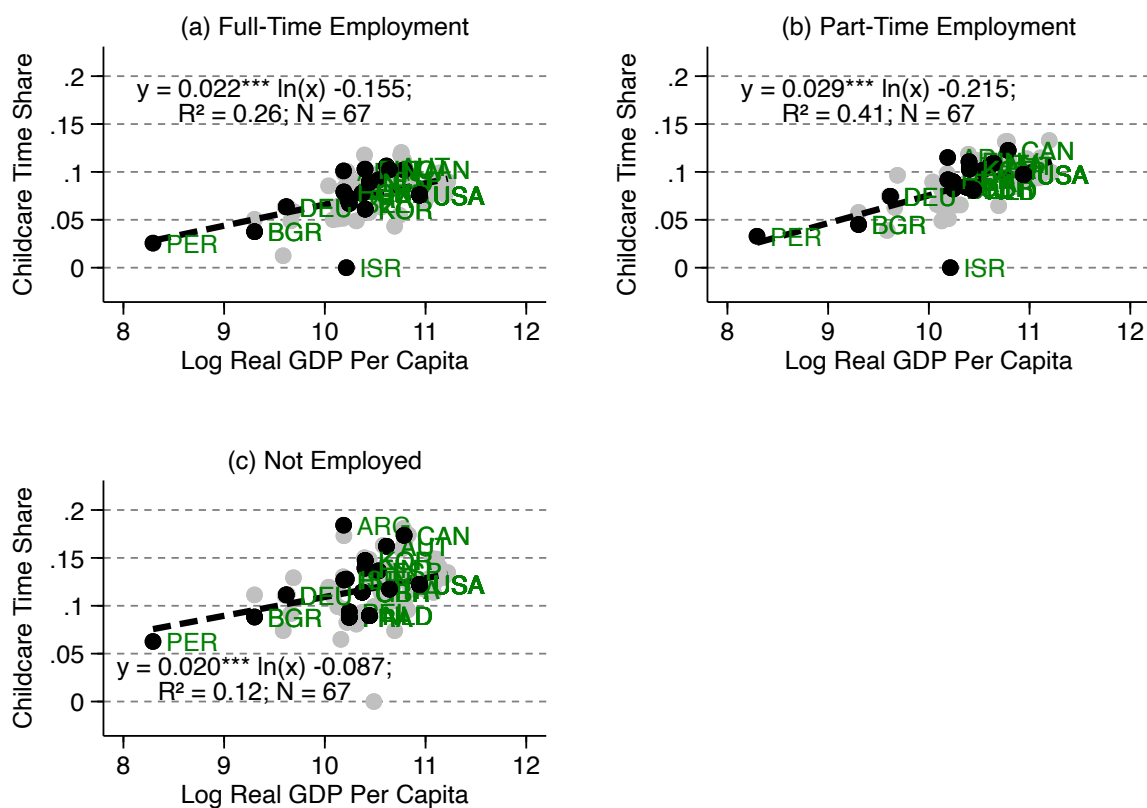


Figure A.5: Female Time Investment in Childcare by Employment Status

Notes: Panel (a) plots the time share allocated to the childcare of young children (ages 0–4) by females in full-time employment. Panel (b) plots the time share allocated to the childcare of young children (ages 0–4) by females in part-time employment. Panel (c) plots the time share allocated to the childcare of young children (ages 0–4) by females that are not employed. Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed under-5 child survey weights. The dashed line corresponds to the line of best fit for the grey circles. The OLS model regresses the parental time investment in child index on log PPP-adjusted real GDP per capita using the country-year observations. *(**)[***] statistically significant at the 10% (5%) [1%] level.

B Model Derivations

B.1 Firm problem

The L -sector producer selects the quantities of $S_L(t)$ and $N_L(t)$ to maximize profits given wages

$$\max_{S_L(t), N_L(t)} \pi_L(t) = p_L(t)A_L(t) \left[\phi_L N_L(t)^{\frac{\epsilon-1}{\epsilon}} + (1 - \phi_L) S_L(t)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}} - w_{L,S}(t)S_L(t) - w_{L,N}(t)N_L(t). \quad (\text{B.1.1})$$

The first order conditions imply

$$(1 - \phi_L) p_L(t) A_L(t) \left(\frac{Z_L(t)}{S_L(t)} \right)^{\frac{1}{\epsilon}} = w_{L,S}(t) \quad \text{and} \quad (\text{B.1.2})$$

$$\phi_L p_L(t) A_L(t) \left(\frac{Z_L(t)}{N_L(t)} \right)^{\frac{1}{\epsilon}} = w_{L,N}(t). \quad (\text{B.1.3})$$

Taking the ratio of (B.1.2) and (B.1.3) and cancelling out $Z_L(t)$ gives

$$\frac{N_L(t)}{S_L(t)} = \left(\frac{(1 - \phi_L) w_{L,S}(t)}{\phi_L w_{L,N}(t)} \right)^{\epsilon}. \quad (\text{B.1.4})$$

The G -sector chooses the quantity of $N_G(t)$ to maximize profits given wages

$$\max_{N_G(t)} \pi_G(t) = p_G(t)A_G(t)N_G(t) - w_{G,N}(t)N_G(t). \quad (\text{B.1.5})$$

The first order conditions imply

$$p_G(t)A_G(t) = w_{G,N}(t). \quad (\text{B.1.6})$$

Finally, the H -sector producer chooses the quantity of skilled labor to maximize profits given wages

$$\max_{S_H(t)} \pi_H(t) = p_H(t)A_H(t)S_H(t) - w_{H,S}(t)S_H(t). \quad (\text{B.1.7})$$

The first order conditions imply

$$p_H(t)A_H(t) = w_{H,S}(t). \quad (\text{B.1.8})$$

B.2 Proof for Proposition 1

I drop the generation superscript and dynasty subscript for expositional convenience. Assume a fixed parent human capital level, prices and wages. Let $V^*(y(t); t)$ denote the value of the parent's problem after maximizing utility conditional on gross income $y(t)$. Furthermore, let $\tau^*(t)$ denote the corresponding optimal time allocation to childcare with a gross income of $y(t)$. The parent's objective function is

$$V^*(y(t); t) = \log\left(\frac{y(t)(1 - \tau^*(t))}{p(t)}\right) - \omega \frac{(1 - \tau^*(t))^{1 + \frac{1}{\chi}}}{1 + \frac{1}{\chi}} + \Phi(\tau^*; t)$$

where $\Phi(\tau^*; t)$ captures the continuation value consisting of subsequent flow utilities and the present discounted value of future generations' utility under the optimal time allocation $\tau^*(t)$.

Consider now a higher gross income $y'(t) > y(t)$. The optimal time allocation $\tau^*(t)$ remains feasible and the continuation value $\Phi(\tau^*; t)$ is unchanged. Now, however, the parent can benefit from higher utility from consumption since

$$\log\left(\frac{y'(t)(1 - \tau^*(t))}{p(t)}\right) - \omega \frac{(1 - \tau^*(t))^{1 + \frac{1}{\chi}}}{1 + \frac{1}{\chi}} > \log\left(\frac{y(t)(1 - \tau^*(t))}{p(t)}\right) - \omega \frac{(1 - \tau^*(t))^{1 + \frac{1}{\chi}}}{1 + \frac{1}{\chi}}$$

It follows that $V^*(y'(t); t) > V^*(y(t); t)$. Accordingly, the parent's intertemporal utility is increasing in gross income.

B.3 Obtaining Marshallian Demands

Given optimal time allocation, each household chooses the consumption bundle that minimizes total expenditure on consumption given the non-homothetic CES aggregator $\forall t \in [gT, (g+1)T)$

$$\min_{\{c_{i,j}^g(t)\}_{t \in [gT, (g+1)T), \forall j \in \{G,L,H\}}} \sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t) \quad \text{subject to} \quad \sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (\text{B.3.1})$$

The corresponding Lagrangian is

$$\mathcal{L} = \sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t) + \Lambda(t) \left[1 - \sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} \right], \quad (\text{B.3.2})$$

where $\Lambda_j(t)$ is the Lagrange multiplier. The corresponding first order condition (FOC) is

$$p_j(t) = \Lambda(t) \frac{\sigma-1}{\sigma} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{-1}{\sigma}} \frac{1}{c_i^g(t)^{\lambda_j}}. \quad (\text{B.3.3})$$

Multiplying both sides by the sectoral good $c_{i,j}^g$ implies

$$p_j(t) c_{i,j}^g(t) = \Lambda(t) \frac{\sigma-1}{\sigma} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}}. \quad (\text{B.3.4})$$

To get rid of the Lagrange multiplier, sum (B.3.4) over all j and plug in the non-homothetic CES aggregator

$$\sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t) = \Lambda(t) \frac{\sigma-1}{\sigma}. \quad (\text{B.3.5})$$

Defining $p_i^g(t) c_i^g(t) = \sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t)$, (B.3.5) can be substituted into (B.3.3) to obtain

$$p_j(t) = p_i^g(t) c_i^g(t) \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{-1}{\sigma}} \frac{1}{c_i^g(t)^{\lambda_j}}. \quad (\text{B.3.6})$$

Rearranging (B.3.6) to solve for the sectoral good yields

$$c_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)c_i^g(t)} \right)^{-\sigma} c_i^g(t)^{(1-\sigma)\lambda_j}. \quad (\text{B.3.7})$$

While (B.3.7) is indeed the Marshallian demand as it expresses sectoral consumption in terms of total expenditure (which is income in equilibrium), prices and parameters, it can be re-expressed as a function of expenditure shares. To see this, multiply and divide both sides of (B.3.7) by the sectoral price $p_j(t)$ and by total expenditure $p_i^g(t)c_i^g(t)$ to obtain

$$\frac{p_j(t)c_{i,j}^g(t)}{p_i^g(t)c_i^g(t)} = \theta_j [p_i^g(t)c_i^g(t)]^{\sigma-1} p_j(t)^{1-\sigma} c_i^g(t)^{(1-\sigma)\lambda_j}. \quad (\text{B.3.8})$$

Since $p_i^g(t)c_i^g(t) = \sum_{j \in \{G,L,H\}} p_j(t)c_{i,j}^g(t)$, it must be that $\sum_{j \in \{G,L,H\}} \frac{p_j(t)c_{i,j}^g(t)}{p_i^g(t)c_i^g(t)} = 1$ which allows (B.3.8) to take on the interpretation of a share. Accordingly,

$$c_{i,j}^g(t) = e_{i,j}^g(t) \frac{p_i^g(t)c_i^g(t)}{p_j(t)} \quad \text{where} \quad e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \quad (\text{B.3.9})$$

To fully characterize (B.3.9), it is necessary to solve for the ideal price index $p_i^g(t)$. To pin down the price index, sum the expenditure share $e_{i,j}^g(t)$ over all j

$$1 = p_i^g(t)^{\sigma-1} \sum_{j \in \{G,L,H\}} \theta_j p_j(t)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \quad (\text{B.3.10})$$

After re-arranging (B.3.10), the ideal price index $p_i^g(t)$ can be written as

$$p_i^g(t) = \left(\sum_{j \in \{G,L,H\}} \theta_j p_j(t)^{1-\sigma} c_i^g(t)^{(\lambda_j-1)(1-\sigma)} \right)^{\frac{1}{1-\sigma}}. \quad (\text{B.3.11})$$

B.4 Proof of Proposition 2

Pre-requisites. The proof requires two ingredients. The first is the sectoral expenditure shares of sector $j \in \{G, L, H\}$ in equilibrium

$$e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \quad (\text{B.4.1})$$

The second is the time derivative of the free labor mobility conditions of skilled and unskilled work. Recall that the free labor mobility conditions, equations (16) and (17), are

$$p_G(t)A_G(t) = \phi_L p_L(t)A_L(t) \left(\frac{Z_L(t)}{N_L(t)} \right)^{\frac{1}{\epsilon}} \quad \text{and} \quad (\text{B.4.2})$$

$$A_H(t) = (1 - \phi_L) p_L(t)A_L(t) \left(\frac{Z_L(t)}{S_L(t)} \right)^{\frac{1}{\epsilon}}, \quad (\text{B.4.3})$$

respectively. Starting with the skilled work free labor mobility condition (B.4.3), taking logs yields

$$\log(A_H(t)) = \log(1 - \phi_L) + \log(p_L(t)) + \log(A_L(t)) + \frac{1}{\epsilon} (\log(Z_L(t)) - \log(S_L(t))). \quad (\text{B.4.4})$$

Then, taking the time derivative of (B.4.4) and rearranging generates

$$\frac{\dot{p}_L(t)}{p_L(t)} = \frac{\dot{A}_H(t)}{A_H(t)} - \frac{\dot{A}_L(t)}{A_L(t)} - \frac{1}{\epsilon} \left(\frac{\dot{Z}_L(t)}{Z_L(t)} - \frac{\dot{S}_L(t)}{S_L(t)} \right). \quad (\text{B.4.5})$$

Repeating for unskilled work free labor mobility (B.4.2) yields

$$\frac{\dot{p}_G(t)}{p_G(t)} = \frac{\dot{p}_L(t)}{p_L(t)} + \frac{\dot{A}_L(t)}{A_L(t)} - \frac{\dot{A}_G(t)}{A_G(t)} + \frac{1}{\epsilon} \left(\frac{\dot{Z}_L(t)}{Z_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right). \quad (\text{B.4.6})$$

Replacing (B.4.5) into (B.4.6) yields

$$\frac{\dot{p}_G(t)}{p_G(t)} = \frac{\dot{A}_H(t)}{A_H(t)} - \frac{\dot{A}_G(t)}{A_G(t)} + \frac{1}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right). \quad (\text{B.4.7})$$

Together, (B.4.5) and (B.4.7) characterize the evolution of relative prices using the two free labor mobility conditions. Notice that these depend on the labor allocations and the TFP growth rates.

Main proof. For the H -sector to dominate, it must be that $R_k(t) \rightarrow 0$ as $t \rightarrow \infty \forall k \in \{L, G\}$ where $R_k(t)$ is the ratio of sectoral expenditure shares

$$R_k(t) = \frac{\theta_k p_k(t)^{1-\sigma} c_i^g(t)^{(\lambda_k-1)(1-\sigma)}}{\theta_H p_H(t)^{1-\sigma} c_i^g(t)^{(\lambda_H-1)(1-\sigma)}}. \quad (\text{B.4.8})$$

Taking logs of $R_k(t)$ yields

$$\log(R_k) = \log\left(\frac{\theta_k}{\theta_H}\right) + (1-\sigma)(\log(p_k(t)) - \log(p_H(t))) + (1-\sigma)(\lambda_k - \lambda_H)\log(c_i^g(t)). \quad (\text{B.4.9})$$

The time derivative becomes

$$\frac{\dot{R}_k(t)}{R_k(t)} = (1-\sigma)\left(\frac{\dot{p}_k(t)}{p_k(t)} - \frac{\dot{p}_H(t)}{p_H(t)}\right) + (\lambda_k - \lambda_H)(1-\sigma)\frac{\dot{c}_i^g(t)}{c_i^g(t)}. \quad (\text{B.4.10})$$

Since $p_H(t)$ is the numéraire, $\frac{\dot{p}_H(t)}{p_H(t)} = 0$. As a result,

$$\frac{\dot{R}_k(t)}{R_k(t)} = (1-\sigma)\left(\frac{\dot{p}_k(t)}{p_k(t)}\right) + (\lambda_k - \lambda_H)(1-\sigma)\frac{\dot{c}_i^g(t)}{c_i^g(t)}. \quad (\text{B.4.11})$$

Intuitively, (B.4.11) shows that the ratio of sectoral expenditure shares depends on how relative prices and income effects evolve. With gross complementarities in sectoral consumption (i.e. $\sigma < 1$), a decline in the relative price of sector k is one force in favor of the H -sector dominating as the k -sector would mechanically occupy a smaller share of expenditure. Another force is income effects: $\lambda_k < \lambda_H$ would imply that the H -sector is income elastic relative to the k -sector good, implying that the household biases consumption towards that sector provided that consumption growth is positive (i.e. $\frac{\dot{c}_i^g(t)}{c_i^g(t)} > 0$).

Next, substituting relative price growth rates (B.4.5) and (B.4.7) into (B.4.11) $\forall k \in \{L, G\}$ and recognizing that $g_j = \frac{\dot{A}_j(t)}{A_j(t)}$, I obtain

$$\frac{\dot{R}_L(t)}{R_L(t)} = \underbrace{(1-\sigma)(g_H - g_L) + \frac{(1-\sigma)}{\epsilon}\left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{Z}_L(t)}{Z_L(t)}\right)}_{\text{relative price effects}} + \underbrace{(\lambda_L - \lambda_H)(1-\sigma)\frac{\dot{c}_i^g(t)}{c_i^g(t)}}_{\text{income effects}}, \quad \text{and} \quad (\text{B.4.12})$$

$$\frac{\dot{R}_G(t)}{R_G(t)} = \underbrace{(1-\sigma)(g_H - g_G) + \frac{(1-\sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right)}_{\text{relative price effects}} + \underbrace{(\lambda_G - \lambda_H)(1-\sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}}_{\text{income effects}}. \quad (\text{B.4.13})$$

Equations (B.4.12) and (B.4.13) reveal that the relative price and income effects are influenced by a combination of parameters and labor allocations. The parametric assumptions of this proposition assume that $0 < g_H < g_L, g_G$ and that $0 < \sigma < 1$. That is, TFP growth is slowest in H -sector, a force inducing relative prices in L - and G -sectors to decline over time since all three sectoral goods are complements (i.e. $0 < \sigma < 1$). Relative prices are also influenced by the evolution of labor allocations, although this effect is modulated by the elasticity of substitution between skilled and unskilled work in L -sector production, ϵ . Intuitively, ϵ controls the effect of different skill types in L -sector production on relative prices. With $\epsilon = \infty$, skilled and unskilled labor are perfect substitutes and so the skill mix in L -sector production does not affect the marginal product of labor and, by extension, relative prices so relative prices are driven solely by the TFP growth rates and elasticity of substitution in consumption between different sectors.

Income effects, on the other hand, are governed by the income elasticity parameters: the parametric assumption that $\lambda_H > \lambda_G, \lambda_L > 0$ together with the assumption of gross complementarities in consumption imply that H -sector is the income elastic sector. Consequently, households bias their consumption to H -sector and away from L - and G -sectors with income.

The parametric assumptions of this proposition imply that the following holds $\forall t$:

$$-x_G > (1-\sigma)(g_H - g_G) + (\lambda_G - \lambda_H)(1-\sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}, \quad \text{and} \quad (\text{B.4.14})$$

$$-x_L > (1-\sigma)(g_H - g_L) + (\lambda_L - \lambda_H)(1-\sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}. \quad (\text{B.4.15})$$

Equations (B.4.14) and (B.4.15) simply say that the cumulative effect of the parameter restrictions is a net negative force on both $R_L(t)$ and $R_G(t)$, respectively.

Let $\Delta_L(t) = \frac{(1-\sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{Z}_L(t)}{Z_L(t)} \right)$ and $\Delta_G(t) = \frac{(1-\sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right)$. To proceed, I use the fundamental theorem of calculus. Denote $f_k(t) = \log(R_k(t))$ and $\dot{f}_k(t) = \frac{\dot{R}_k(t)}{R_k(t)} \quad \forall k \in \{L, G\}$. Combining (B.4.12)-(B.4.15) with the definition of $\Delta_k(t)$ gives

$$\dot{f}_k(t) < -x_k + \Delta_k(t) \quad \forall k \in \{L, G\}. \quad (\text{B.4.16})$$

A result of the fundamental theorem of calculus is that

$$f_k(t) - f_k(0) = \int_0^t \dot{f}_k(s) ds. \quad (\text{B.4.17})$$

Replacing (B.4.16) into (B.4.17) and integrating it out yields

$$f_k(t) - f_k(0) < -x_k t + \int_0^t \Delta_k(s) ds. \quad (\text{B.4.18})$$

That is, a sufficient condition for $f_k(t) \rightarrow -\infty$ is that the linear term $-x_k t$ dominate $\int_0^t \Delta_k(s) ds$ as $t \rightarrow \infty$. Dividing (B.4.18) by t , this holds whenever

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t \Delta_k(s) ds < x_k. \quad (\text{B.4.19})$$

Condition (B.4.19) is the additional requirement in the $\epsilon < \infty$ case. Since $\int_0^t \Delta_k(s) ds = \frac{1-\sigma}{\epsilon} \left[\log \frac{S_L(t)}{S_L(0)} - \log \frac{X_L^k(t)}{X_L^k(0)} \right]$, it would hold whenever the labor allocations remain bounded away from 0 and ∞ , since then $\frac{1}{t} \int_0^t \Delta_k(s) ds \rightarrow 0 < x_k$; but because the allocation growth rates $\frac{\dot{S}_L}{S_L}$, $\frac{\dot{Z}_L(t)}{Z_L(t)}$, and $\frac{\dot{N}_L(t)}{N_L(t)}$ have no closed form, (B.4.19) is verified numerically along the transition path. In the perfect-substitutes case $\epsilon = \infty$, the factor $\frac{1-\sigma}{\epsilon}$ vanishes, so $\Delta_k(t) \equiv 0$ and (B.4.18) reduces to

$$f_k(t) - f_k(0) < -x_k t, \quad (\text{B.4.20})$$

where it is easy to see that since x_k is a constant, as $t \rightarrow \infty$, $f_k(t) \rightarrow -\infty$. With $R_k(t) = \exp(f_k(t))$, this means that $\exp(-\infty) = 0$ which means that the expenditure share of H -sector asymptotically dominates that of the L - and G -sectors.

C Quantitative Analysis

C.1 Numerical Solution

To solve the model numerically, I adapt the algorithm in Buera and Shin (2017). The algorithm is composed of two parts. The first part computes the steady state value function and distribution based on a stationarized, one-sector version of the where a single sector dominates (as per Section 3.4). The second part computes the transition by iterating over price sequences until convergence using the steady state as the terminal condition.

The state space is 2-dimensional, defined on parent-child human capital pairs. In recognition that the support for initial childhood human capital grids may differ from the parent human capital support, I distinguish between $\mathcal{H}$ and $\mathcal{H}'_0$ where the former is the support for parent human capital and the latter is the support for child human capital obtained from the initial draws. The state space is therefore defined along the $\mathcal{H} \times \mathcal{H}'_0$ dimensions.

To proceed, I discretize the parent human capital support $\mathcal{H}_d = \{h_1, \dots, h_K\}$ and the child human capital draw support $\mathcal{H}'_{0,d} = \{h'_{0,1}, \dots, h'_{0,K}\}$. I also discretize the initial child human capital draw probability density function $g(\eta_c, \kappa_c)$ into a probability mass function $g_d(\eta_c, \kappa_c)$. Finally, I generate sequences of $A_j(t)$ which are constructed by feeding the constant sectoral TFP growth rates g_j into a compound growth function.

Steady State. Algorithm 1 presents the solution method for solving for the steady state in a single-sector economy with only H -sector output being consumed and produced. This is done in three steps. The first step performs value function iteration over the generation's problem for each parent human capital-child draw pairs. This step pins down the optimal policy function for each pair and the terminal condition, which is the value function in steady state V^{ss} . The second step iterates over the parent human capital distribution using the optimal time allocation policy function and the initial child human capital draws. The third step ensures that the market for H -sector output clears. Both consumption and skills supply are aggregated using the parent human capital and initial child human capital distributions as weights.

Algorithm 1 Steady-State Equilibrium

Require: Grids $\mathcal{H}_d, \mathcal{H}'_{0,d}$, child pmf $g_d(\eta_c, \kappa_c)$, choice set $\mathbb{T}^S$, discount factors β, β_g , tolerance ε , and maximum iterations $i_{\max}$.

Ensure: $V^{\text{ss}}, \mu^{\text{ss}}$, and τ^{ss} .

- 1: Stationarize wages: $w(t)/A(t) = 1$ for all $t > T$.
- 2: Initialize $V^{(1)}$, set $i \leftarrow 1$ and $\Delta_V \leftarrow \infty$.
- 3: **while** $\Delta_V > \varepsilon$ and $i < i_{\max}$ **do**
- 4: **for** $(h, h'_0) \in \mathcal{H}_d \times \mathcal{H}'_{0,d}$ **do**
- 5: Solve

$$\tilde{V}(h, h'_0) = \max_{\tau \in \mathbb{T}^S} \left\{ \sum_{s=1}^S \beta^{s-1} u(c_s, \tau_s) + \beta^T \beta_g \mathcal{I}[V^{(i)}](h'(h, h'_0, \tau)) \right\}, \quad c_s = (1 - \tau_s)h,$$

and store $\tau^{\text{ss}}(h, h'_0)$ and $c^{\text{ss}}(h, h'_0)$.

- 6: **end for**
 - 7: $V^{(i+1)}(h) \leftarrow \sum_{h'_0 \in \mathcal{H}'_0} g_d(h'_0) \tilde{V}(h, h'_0)$.
 - 8: $\Delta_V \leftarrow \max_{h \in \mathcal{H}} |V^{(i+1)}(h) - V^{(i)}(h)|$; $i \leftarrow i + 1$.
 - 9: **end while**
 - 10: $V^{\text{ss}} \leftarrow V^{(i)}$.
 - 11: Construct Π^{ss} from τ^{ss} and iterate $\mu^{(i+1)} = \Pi^{\text{ss}} \mu^{(i)}$ until convergence; set $\mu^{\text{ss}} \leftarrow \mu^{(i)}$.
 - 12: **for** $s = 1, \dots, S$ **do**
 - 13: $c_s^{\text{ss}} \leftarrow \mathcal{L} \sum_{h, h'_0} \mu^{\text{ss}}(h) g_d(h'_0) c_s^{\text{ss}}(h, h'_0)$.
 - 14: $S_s^{\text{ss}} \leftarrow \mathcal{L} \sum_{h, h'_0} \mu^{\text{ss}}(h) g_d(h'_0) h (1 - \tau_s^{\text{ss}}(h, h'_0))$.
 - 15: Check $|c_s^{\text{ss}} - S_s^{\text{ss}}| < \varepsilon$.
 - 16: **end for**
-

Transition. Algorithm 2 presents the solution method applied to solve for the transition. Crucially, this is achieved by iterating over the fixed point sequences of L - and G -sector prices until markets clear and free labor mobility holds.

Algorithm 3 is called in line 7. This algorithm takes as inputs the proposed price paths, solves the household problem and computes the joint distribution and, accordingly, the aggregate quantities $c_j(t), S(t), N(t) \forall t < T, j \in \{G, H, L\}$. This is done in three steps. The first step takes as input the steady state value function solved in Algorithm 1 and, through backward induction, computes optimal policy for each household human capital pairs. The second step uses the optimal time allocation policy and shoots forward using the Kolmogorov Forward Equation (KFE) to solve for the joint distribution in all generations. The third step uses the joint distribution to compute equilibrium aggregates including the supply of skills in the economy and sectoral output demand given the proposed price paths.

Algorithm 2 Transition-Path Equilibrium: Price Block

Require: TFP paths $\{A_j(t)\}_{t \in \mathcal{D}}$, $j \in \{G, H, L\}$, terminal value V^{ss} , initial distribution μ_1 , damping parameters $\gamma, \xi_{\text{lo}}, \xi_{\text{hi}}$, tolerances $\varepsilon_L, \varepsilon_G$, and iteration limits.

Ensure: Equilibrium paths $\{p_G(t), p_L(t)\}_{t \in \mathcal{D}}$.

- 1: Set $p_H(t) \leftarrow 1$ for all $t \in \mathcal{D}$. ▷ Numeraire
 - 2: Initialize $p_G^{(1)}, i_G \leftarrow 1$, and $\Delta_G \leftarrow \infty$.
 - 3: **while** $\Delta_G > \varepsilon_G$ and $i_G \leq i_{G,\text{max}}$ **do**
 - 4: Compute $\bar{h}(t)$ for all $t \in \mathcal{D}$.
 - 5: Initialize $p_L^{(1)}, i_L \leftarrow 1$, and $\Delta_L \leftarrow \infty$.
 - 6: **while** $\Delta_L > \varepsilon_L$ and $i_L \leq i_{L,\text{max}}$ **do**
 - 7: Obtain $c_j(t)$, $S(t)$, and $N(t)$ from Algorithm 3. Back out $N_j(t)$ and $S_j(t)$.
 - 8: Compute implied prices $p_L^{\text{imp}}(t)$.
 - 9: $\Delta_L \leftarrow \max_{t \in \mathcal{D}} |p_L^{(i_L)}(t)/p_L^{\text{imp}}(t) - 1|$.
 - 10: $p_L^{(i_L+1)}(t) \leftarrow p_L^{(i_L)}(t) - \gamma[p_L^{(i_L)}(t) - p_L^{\text{imp}}(t)], t \in \mathcal{D}$.
 - 11: $i_L \leftarrow i_L + 1$.
 - 12: **end while**
 - 13: Compute implied prices $p_G^{\text{imp}}(t)$ and errors $e_G(t) \leftarrow p_G^{(i_G)}(t)/p_G^{\text{imp}}(t) - 1$.
 - 14: $\Delta_G \leftarrow \max_{t \in \mathcal{D}} |e_G(t)|$.
 - 15: Set $\xi(t) \leftarrow \xi_{\text{lo}}$ if $|e_G(t)| < 0.3$ and ξ_{hi} otherwise.
 - 16: $p_G^{(i_G+1)}(t) \leftarrow p_G^{(i_G)}(t) - \xi(t)[p_G^{(i_G)}(t) - p_G^{\text{imp}}(t)], t \in \mathcal{D}$.
 - 17: $i_G \leftarrow i_G + 1$.
 - 18: **end while**
-

Algorithm 3 Transition-Path Equilibrium: Household and Distribution Block

Require: Candidate paths $\{p_G(t), p_L(t), \bar{h}(t)\}_{t \in \mathcal{D}}$, terminal value V^{ss} , initial distribution μ_1 , grids $\mathcal{H}_d, \mathcal{H}'_{0,d}$, child pmf $g_d(\eta_c, \kappa_c)$, choice set $\mathbb{T}^S$, and discount factors β, β_g .

Ensure: $\{c_j(t), S(t), N(t)\}_{t \in \mathcal{D}}, j \in \{G, H, L\}$.

```

1: Set  $\bar{V}_G(h) \leftarrow V^{\text{ss}}(h)$ .
2: for  $g = G - 1$  down to 1 do
3:   for  $h \in \mathcal{H}_d$  do
4:     for  $h'_0 \in \mathcal{H}'_{0,d}$  do
5:       for  $\tau \in \mathbb{T}^S$  do
6:         for  $s = 1, \dots, S$  do
7:           Set  $t \leftarrow (g - 1)S + s$ .
8:           Determine  $o^*(h; t)$  and gross income  $w_s(h; t)$ .
9:           Set  $c_s(h, h'_0; g) \leftarrow (1 - \tau_s)w_s(h; t)$ .
10:        end for
11:         $Q_g(h, h'_0, \tau) \leftarrow \sum_{s=1}^S \beta^{s-1} u(c_s, \tau_s) + \beta^T \beta_g \mathcal{I}[\bar{V}_{g+1}](h'(h, h'_0, \tau))$ .
12:      end for
13:       $\tau^*(h, h'_0; g) \leftarrow \arg \max_{\tau \in \mathbb{T}^S} Q_g(h, h'_0, \tau)$ .
14:      Store  $c^*(h, h'_0; g)$  and  $V_g(h, h'_0) \leftarrow \max_{\tau \in \mathbb{T}^S} Q_g(h, h'_0, \tau)$ .
15:    end for
16:     $\bar{V}_g(h) \leftarrow \sum_{h'_0 \in \mathcal{H}'_{0,d}} g_d(h'_0) V_g(h, h'_0)$ .
17:  end for
18: end for
19: Initialize  $\mu_1(h, h'_0) \leftarrow \mu_1(h) g_d(h'_0)$ .
20: for  $g = 1, \dots, G - 2$  do
21:   Propagate  $\mu_g$  through  $h'(h, h'_0, \tau^*(h, h'_0; g))$  to obtain  $\mu_{g+1}(h)$ .
22:   Set  $\mu_{g+1}(h, h'_0) \leftarrow \mu_{g+1}(h) g_d(h'_0)$ .
23: end for
24: for  $t \in \mathcal{D}$  do
25:   Set  $g \leftarrow \lceil t/S \rceil$  and  $s \leftarrow t - (g - 1)S$ .
26:    $S(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) h (1 - \tau_s^*) \mathbf{1}\{h > \bar{h}(t)\}$ .
27:    $N(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) (1 - \tau_s^*) \mathbf{1}\{h \leq \bar{h}(t)\}$ .
28:    $c_j(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) c_{j,s}^*(h, h'_0; g), j \in \{G, H, L\}$ .
29: end for

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C.2 The Timing Convention

Table C.1 connects generations and stages of childhood development that are used to build the synthetic cohorts for the time use data with calendar years. That way, the generation-level moments such as skilled worker and expenditure shares are 15-year averages over the same calendar years that define a generation. This ensures consistency with the stage-specific childcare time moments which partition that window into three 5-year stages of childhood development.

Generation	Stage of development	Calendar years	Age of child (years, inclusive)
1	1	1942–1946	0–4
	2	1947–1951	5–9
	3	1952–1956	10–14
2	1	1957–1961	0–4
	2	1962–1966	5–9
	3	1967–1971	10–14
3	1	1972–1976	0–4
	2	1977–1981	5–9
	3	1982–1986	10–14
4	1	1987–1991	0–4
	2	1992–1996	5–9
	3	1997–2001	10–14
5	1	2002–2006	0–4
	2	2007–2011	5–9
	3	2012–2016	10–14
6	1	2017–2021	0–4
	2	2022–2026	5–9
	3	2027–2031	10–14

Table C.1: Linking Generations, Stages of Childhood Development, and Calendar Years

Notes: This table maps each model generation and stage of childhood development into the corresponding calendar years and age range. Each generation contains three five-year stages of childhood development: stage 1 covers ages 0–4, stage 2 covers ages 5–9, and stage 3 covers ages 10–14. The mapping is used to identify generation- and stage-specific empirical moments and to construct the synthetic cohorts underlying the parental time-allocation moments.

C.3 Constructing Targeted Data Moments

Time Allocation. I obtain the time allocation moments from the Multinational Time Use Survey (Lamote de Grignon Perez et al., 2024), restricting attention to the US samples which are available from 1965 onward. I keep individuals who are (i) in full-time employment, (ii) between the ages of 21 and 64 (inclusive) and (iii) live in a household that contains a child below the age of 15.²³ To remain consistent with the model, the parental childcare time shares are defined as the time allocated to childcare divided by the time allocated to both childcare and paid work.

Since there exists a single cohort of both parents and children at any given time in the model, all children in the economy are in the same stage of childhood development. I therefore proceed by reconstructing time allocation profiles for synthetic cohorts using age-restricted cross sections. Tracking the time allocation of parents of a given cohort over time (using the timing convention Appendix C.2) enables identification of stage-specific elasticities of parental time allocation. Where multiple calendar years are available for a given parent generation-child stage of development period, I take a simple average. Diarist-day observations for each sample are weighted using the proposed survey weights. The five time allocation moments are:

Generation 3, stage 1 (1972-1976)

- Time allocated to childcare for diarists who live in a household with a child between 0-4 years old

Generation 6, stage 1 (2017-2021)

- Time allocated to childcare for diarists who live in a household with a child between 0-4 years old

Generation 4, stage 2 (1992-1996)

- Time allocated to childcare for diarists who live in a household with a child between 5-9 years old

Generation 6, stage 2 (2022-2026)

- Time allocated to childcare for diarists who live in a household with a child between 5-9 years old

Generation 3, stage 3 (1982-1986)

- Time allocated to childcare for diarists who live in a household with a child between 10-14 years old

²³The surveys do not capture how the diarist relates to the child.

Skill Distribution. The skill distribution data moments are computed using microdata obtained from IPUMS USA (Ruggles et al., 2025). Attention is restricted to individuals who are (i) employed, (ii) between the ages of 21 and 64 (inclusive), (iii) work at least 30 hours a week and (iv) worked for at least 40 weeks.²⁴ Skilled workers are defined as workers whose occupation is either managerial, professional or technical (Acemoglu and Autor, 2011), where the occupation identifier variable is OCC1950. Skills are assumed fixed after childhood, meaning that the skill distribution is a generation-specific distribution. Thus, where multiple sample calendar years are available for a single generation, I take a simple average. All skill distribution moments are weighted using person weights. The five targeted skill distribution moments are:

Generation 1, 1942-1956

- Share of skilled workers
- Log 90-10 wage per hour ratio of skilled workers

Generation 3, 1972-1986

- Share of skilled workers

Generation 6, 2017-present

- Share of skilled workers
- Log 90-10 wage per hour ratio of skilled workers

Structural Change. The production function does not feature intermediate inputs. Thus, to consistently map consumption expenditures to the data, it is necessary to extract the consumption component of total value added at the broad sector level. Herrendorf et al. (2013) lay out the procedure to do this in their Online Appendix. For the sake of completeness, I provide a brief summary of these steps in Appendix C.4 together with the time series of consumption value added for my three broad sectors of interest. The procedure requires input-output tables and the total requirements matrix which are made available by the BEA for benchmark years (U.S. Bureau of Economic Analysis, 2025). Where data availability permits, I average over the 15-window for a given generation in order to

²⁴Hours worked last week (i.e. HRSWORK1) is not available for all samples. Where unavailable, I proxy hours worked last week by usual hours worked per week (i.e. UHRSWORK). Hours worked per year are proxied by the product of HRSWORK1 and WKSWORK1 where available and UHRSWORK and WKSWORK1 otherwise.

remove stage-specific trends and isolate cross-sectional variation in sectoral expenditure shares. The four targeted structural change moments are:

Generation 1, 1942-1956

- High-Skill Services expenditure share
- Goods expenditure share

Generation 6, 2017-present

- High-Skill Services expenditure share
- Goods expenditure share

C.4 Extracting Consumption Value Added

Herrendorf et al. (2013) offer a complete description of the steps required to extract sectoral consumption value added from input-output tables in the Online Appendix B. Given that I distinguish between High- and Low-Skill Services and aggregate Agriculture and Manufacturing into Goods, I repeat this exercise using the same input-output tables but with the relevant sectoral split. I provide a brief description of the steps below for completeness.

Let m be the number of commodities produced by the economy and n be the number of industries that produce these m commodities. In input-output tables prior to 1972, $m = n$.

1. From the Use table, extract the final consumption expenditure vector $\mathbf{e}_c$. This is an $m \times 1$ vector where each element i denotes the value of final consumption expenditure spent on the corresponding commodity.
2. From the Use table, extract total output and total value added by industry. Compute the value added share of output by taking the ratio of total value added over total output for each industry. This vector $\mathbf{v}$ is an $n \times 1$ vector.
3. Extract the total requirements matrix $\mathbf{R}$. In post-1972 tables, this is an $n \times m$ matrix. Entry ji of the total requirements matrix gives the dollar value of industry j 's production that is required to deliver one dollar of the domestically produced commodity i to final uses.
4. Consumption value added by industry is then

$$\mathbf{c} = \langle \mathbf{v} \rangle \mathbf{R} \mathbf{e}_c$$

where $\langle \mathbf{v} \rangle$ is the diagonal matrix with vector $\mathbf{v}$ on its diagonal. Element i of $\mathbf{c}$ denotes the consumption value added of the corresponding industry.

5. Aggregate industries into sectors by summing the elements of $\mathbf{c}$ corresponding to each sector as per the industry classification in Section 2. Repeat these steps for each year where input-output tables are available.

These steps are performed using the BEA's Historical Benchmark Input-Output Tables for years where data is available (1947, 1958, 1963, 1972, 1977, 1987, 1992) as well as the annual Input-Output Accounts Data releases for later years (1997-2024).